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05	Gap Analysis	□□□□□□□□5□□□□□	□□□□□□□□□□□□□□
06	Architecture Proposal	□□□□□□□□□	□□□□□□LangGraph, Veo 3.1, LangSmith
07	Workflow Analysis	□□□□□□□□□□□□ (2026.03)	□□□□□2 □□□□□Kling/Seedance/Veo □□□□□ C □□
08	Best Practices	**AI** □□□□□□□□□□□□	4 □□□□□□□□6 □□□□□□□□□□□□□□Prompt □□
09	Commercial Production	□□□□ **AI** □□□□□□□□ □□	□□□□□□Runway □□□□□□□□□□□□□□□ ~$280
10	**Ultimate Guide**	**AI** □□□□□□□□□□	4 □□□□□8 □□□□□Prompt □□□□□□□□□□□□□□□□□□ □□
11	**One-Click Best Practices**	□□□□□□□□□	GitHub □□□□□□□4 □□□□□□□6 □□□□□□□□□□□□□□ □

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CLI	`src/cli.py`	~310	`videoqa generate/evaluate/script` □□
Orchestrator	`src/pipeline/orchestrator.py`	~400	□□ → □□ → □□ → □□□□
Composer	`src/pipeline/composer.py`	~306	FFmpeg □□□□□□□

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L1 □□□	`src/quality_gate/layer1_programmatic.py`	<1 □	$0	□□□□□□□□□□□□□□□□□□ □□□□□□
L2 □□□□	`src/quality_gate/layer2_ml.py`	1-10□	~$0.01	NIQE □□□□□CLIP □□□□ □□□□□□□□□□
L3 VLM	`src/quality_gate/layer3_vlm.py`	5-30□	~$0.05	Gemini/GPT-4V □□□ CoT □□□□□□□□□□□□ Prompt
Aggregator	`src/quality_gate/aggregator.py`	—	—	□□□□□□□□□□□□

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Diagnosis	`src/correction/diagnosis.py`	5□□□□□ × 3□□□□□□□□□□□□
Prompt Rewriter	`src/correction/prompt_rewriter.py`	VLM □□ + □□□□□□□□□□20+ □□□□□

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AI Video	`src/pipeline/generators/ai_video.py`	**Veo 3.1**, Kling, Runway
Script	`src/pipeline/generators/script.py`	OpenAI, Gemini, fallback templates
TTS	`src/pipeline/generators/tts.py`	edge-tts (□□), ElevenLabs
BGM	`src/pipeline/generators/bgm.py`	□□□ + FFmpeg □□/□□□□
Subtitles	`src/pipeline/generators/subtitles.py`	Whisper (openai-whisper)
Stock Footage	`src/pipeline/generators/stock_footage.py`	Pexels, Pixabay (□□)

Web □□

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FastAPI Server	`src/server.py`	8000	REST API: /api/generate, /api/evaluate, /api/jobs
Gradio Web UI	`src/web_ui.py`	7860	□□□□□TTS□□□□□□□□□

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`src/quality_gate/config.yaml`	L1/L2/L3 □□□□□□□□□□□
`config/quality_thresholds.yaml`	□□□□□/□□□□
`config/pipeline.yaml`	□□□□□□□□□□□□□□□□
`pyproject.toml`	□□□□□□□□□□□□□□

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`tests/test_layer1.py`	11	L1 □□□□□□□□□□□□
`tests/test_layer2.py`	3	L2 □□□□□□□□□□□□□□□
`tests/test_aggregator.py`	5	□□□□□□□
`tests/test_correction.py`	9	□□ + Prompt □□□
`tests/test_orchestrator.py`	6	□□□□□□□□□□□□□□□□□□
`tests/test_generators.py`	24	□□□□□□□□ + □□□

`tests/test_phase5.py`	17	□□□□□□CLI□□□□

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Phase 1	□ □□□	□□□□□ — 6□□□□□
Phase 2	□ □□□	□□□□□ — L1/L2/L3 + □□□
Phase 3	□ □□□	□□□□ — □□□Prompt □□□□□□
Phase 4	□ □□□	□□□□□□ — Veo/Kling/Runway□TTS□BGM□□□□□□□
Phase 5	□ □□□	□□□□ — CLI□FastAPI□Gradio Web UI□Veo 3.1 □□

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VISTA	Google DeepMind	2024	□□□□□□□60% □□□□
Evaluator-Optimizer	Anthropic	2025	□□□□□□□□□□
VBench	Tsinghua	2024	16□□□□□□□□
VMAF	Netflix	2016+	□□□□□□□□
CLIPScore	CMU	2021	□□□□□-□□□□
T2VQA-DB	Multiple	2024	□□□□□□□□□□□
Q-Bench-Video	Multiple	2025	LMM □□□□□□□□ (CVPR 2025)
VGBE Workshop	CVPR	2026	□□□□□□□□□□□□

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Google Veo 3.1	AI □□□□□□□□	$0.15-0.40/□
Kling AI	AI □□□□□□□□	$0.50-1.50/□□
Runway	AI □□□□□□□□	$0.05-0.12/□
edge-tts	□□□□□□□	□□
Pexels API	□□□□□□	□□
OpenAI Whisper	□□□□□□□□□□	□□□□□□
FFmpeg	□□/□□□□	□□

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MIT License — □□ LICENSE

GitHub

https://github.com/joyhpc/ai-short-video-research

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Revideo	~3.7k	TypeScript	Motion Canvas □□□□ □□□□□ API□	□□□□□□□□□□□API □ Remotion □□□□	□□□□□□□□□□□□□

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Sora 2 (OpenAI)

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Digital Human / Avatar

Synthesia

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Jimeng (□□) -- ByteDance

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Kling 3.0 (□□) -- Kuaishou

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Zhipu Qingying (□□□□) -- ZhipuAI

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Hailuo AI (□□AI) -- MiniMax

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□□	**Runway**	**Sora 2**	**Synthesia**	**Kling 3.0**	**Jimeng**	**InVideo**	**Open Source (□□)**
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3. "□□ **10** □□ **1** □"□□□□□□□□ — □□□□ + □□□□□□□□□"□□□□"
4. **87%** □□□□□□□□□□ **Prompt** □□ — □ Prompt □□□□□□□□□
5. □□□□□□□□□**Runway 73%**□□**6** □□□□□□□□□□ **15%** — □□□□□□□□□□
6. **API** □□□□□□□□□□ — □□□□□□□□□□□ 85-95%

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3. □□□□□□□□□□

3.1 □□□□□□□□□

□□□□□□□□□□□□□□□□□□□□

> *□ □□□□□□□□□□□□□□□□□□□□ 54*

3.2 □ 10 □□□□□□□□□□□□□□□

□□	□□/**10s**	□□□□	□□□□/**10s**	**API** □□
Pexels □□	$0	1	**$0**	pexels.com
□□ LTX-2	$0□□ GPU□	1	**$0**	github.com/Lightricks/LTX-2
□□ Wan 2.2	$0□□ GPU□	1	**$0**	github.com/Wan-Video/Wan2.2
Hailuo	$0.19	~2	$0.38	hailuoai.video

Runway Turbo	$0.50	~2	$1.00	runwayml.com
Kling 3.0	$0.60-0.75	~2	$1.20-1.50	klingai.com / ModelsLab
Seedance 2.0	$0.14/s □□	~2	$2.80	□□□□
Sora 2	$1.00	~3	$3.00	sora.com
Veo 3.1 Fast	$1.50	**4-6**	**$6-9**	aistudio.google.com

3.3 □□□□□□□6 □□/□□

□□	□□□□□	**6** □□□□□□□	□□
Pexels+CLIP	45%	0.8%	□□□□□□□□□□
Seedance 2.0	60%	4.7%	20 □□□ 1 □□□□□
Kling 3.0	65%	7.5%	13 □□□ 1 □□□□□
Runway Gen-4	73%	**15.1%**	7 □□□ 1 □□□□□

□□□□□□□□□□□□ **73%**□□□□□**6** □□□□□□□□□□□ **15%**□□□□□□□□□□□□□□□□□□□□□□

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4. □□□□□□□□

4.1 Kling 3.0□□□□— 2026.02.05 □□

> *□ □□□□□□□□□□□□□□□□□□□ **55***

□□	□□
□□□	□□ 2K/4K□30fps□Pro 60fps□
□□□	□ prompt 2-6 □□□□□□□ ~3 □□
□□□□	□□/□□/□□/□□□□□□□□
□□□□□	Subject Reference□□□□□□+ □□□□
□□□□	V3.0 Pro Motion Control□□□□□□□□/□□/□□□
□□	Canvas Agent□□□□+□□□□□□□
□□	Free 66 daily credits; Std $6.99/mo; Pro $25.99/mo; Premier $64.99/mo; Ultra $180/mo
API	ModelsLab ~$0.12-0.15/s; PiAPI $10/mo/seat; □□□□ $4,200/30K units

4.2 Seedance 2.0□□□□□□— 2026.02.12 □□□□

> *□ □□□□□□□□□□□□□□□□□□□ **56***

□□	□□
□□□	□□ 2K

□□□□□	□□ **12** □□□□□□□□/□□/□□□
□□□□	□□□□□□□/□□□/□□/□□□□
□□□□□□□	□/□/□□□□□
□□□□	□□□□□□□□□□□□□/□□
□□	□□□□ ~$0.14/s; Atlas Cloud $0.022/s (v1.5 Pro)
□□	□□ **API** □□□□□□□□□□□□□□□/□□□/MPA□

4.3 Google Veo 3.1

> *□ □□□□□□□□□□□□□□□□□□□ **57***

□□	□□
□□□	720p / 1080p / 4K
□□□□	8 □/□□
□□□□	□□/□□/□□□
□□□□	□□ 9:16
□□	Fast $0.15/s; Standard $0.40/s
□□	Preview 10 RPM; Production 50 RPM
□□	□□□□□□□□□□□□□□ **15-20%**

4.4 Runway Gen-4 / Gen-4.5 — 2026.02.10 API □□

> *□ □□□□□□□□□□□□□□□□□□□ **58***

□□	□□
□□□□□	**73%**□□□□
□□□	□□ 4K
□□	Gen-4 □□ 60s; Gen-4.5 2-10s
□□□□	□□□□□□□□□□□□□□
□□	Motion Brush 3.0, Director Mode, □□/□□□□□
□□	Adobe Firefly, Envato VideoGen
□□	Free 125 credits; Std $12/mo; Pro $28/mo; Unlimited $76/mo
API	Gen-4 Turbo $0.05/s□□□□□□□ API□

4.5 Sora 2□OpenAI□

> *□ □□□□□□□□□□□□□□□□□□ 59*

□□	□□
□□	□□ 5-15s; Pro □□ 25s□□□□□□
□□□□	□□/□□/□□□□
□□	□□□□□, □□□□, Character Cameos
□□	**2026.01.10** □□□□□□□
□□	ChatGPT Plus $20/mo (1000 credits); Pro $200/mo
API	$0.10/s (base); $0.30/s (Pro 720p); $0.50/s (Pro 1080p)

4.6 Pika 2.5

□□	□□
□□	5-10 □
□□	Pikaffects □□□□□□□/□□/□□/□□□, □□□□□□, □□□□
□□	Free 80 credits; Std $8/mo; Pro $28/mo; Fancy $76/mo
API	fal.ai ~$0.20/5s 720p

4.7 Hailuo / MiniMax

□□	□□
□□	□□□□□□□□□□□AI □□□□□□□□□
□□	Std $9.99/mo; Pro $34.99/mo; Ultra $124.99/mo; Max $199.99/mo
API	$0.19-0.56/video

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5. □□□□□□□□□

5.1 □□□□□□□□

□□	□□	□□□	□□	□□□ **GPU**	□□□	□□
Wan 2.2 (5B)	□□□□	720p@24fps	~5s	□ RTX 3080 (10GB)	MIT	VBench #1, MoE □□
Wan 2.2 (27B)	□□□□	720p@24fps	~5s	□ □ A100 (~60GB)	MIT	□□□□
LTX-2	Lightricks	**4K@50fps**	10-20s	□ RTX 4090 (24GB)	Apache 2.0	□□□□, NVIDIA CES □□

CogVideoX (5B)	□□	720x480	~6s	□ RTX 4090 (24GB)	Apache 2.0	□□□□□□□
HunyuanVideo	□□	720p@24fps	~5s	□ □ A100 (40-80GB)	Open	13B □□, □□□
Mochi 1	Genmo	480p@30fps	~5.4s	□ □ A100 (40-80GB)	Apache 2.0	10B, prompt □□□

□□□ **GPU □□□**

- **RTX 3080/4070 (10-12GB)**□Wan 2.2 (5B) — 720p, ~9 □□/□□
- **RTX 4090 (24GB)**□LTX-2 — □□ 4K, □□□□
- **WanGP** □□□□□□□□ 6GB □□□□□RTX 10XX/20XX, AMD□

5.2 □□□□□□□

□□	**GitHub Stars**	□□□□	□□□□	□□□□
MoneyPrinterTurbo	55k+	Pexels □□□□□	□	v1.2.6 (2025.05)
NarratoAI	~5k	□□□□□□	⚠ VLM □□□□	—
ShortGPT	~5k	Pexels+YouTube	□	—
Short Video Maker	~1k	Pexels □□	□	—
ComfyUI	—	□□ AI □□	□	□□□□

□□□□□□□

□□□□□	30-60 □/□□1-2 □/□□	15-30 □□/□
□□□□	1 □/□	1-2 □□/□
□□□□□□□□	80% AI □□ + 20% □□□□	□□□□□

6.3 □□□□□□

□□	□□□	□□
□□□□	□□	15-30 □□□□
□□□□+□□	□	"□□ 10 □□□ 1 □"
A/B □□	□	□□□□□□□□□□□□
□□□□□□	□	□□□□□□□□

6.4 □□□□□□□

□□	URL	□□
□□ **Dreamina**	jimeng.jianying.com / dreamina.capcut.com	□□□□□□□□□Seedance□
□□ **Kling**	klingai.com	□□ 4K/60fps □□□□
□□ **Hailuo**	hailuoai.video	MiniMax □□□□□□□
□□ **Jianying**	jianying.com	□□□□□□□□□□□□
□□ **Doubao**	doubao.com	□□□□ AI □□□□□ Seedance
WaveSpeedAI	wavespeed.ai	□□□□□□□□600+ □□□
□□ **iClip**	—	RPA □□□□□□/□□/□□/□□□

6.5 □□□□□□□□ (2026 □□□□)

□□□□□□□□□□□□□□□□□□□□ API □□□□□□□□□□□□ "Agentic Workflow"□□□□□□□□□

1. □□□□□□□□□□□□□**ComfyUI**
   - □□□□□□□□□□□□□□□□
   - □□□□□□□ LTX-2, Wan 2.2, □ HunyuanVideo □□□□□□□□□□ `Midjourney□□ -> ComfyUI□□□□ -> ComfyUI□□□□ -> □□□□(Upscale)` □□□□□□□□□□□□□ GPU□24GB+ VRAM□□□□□□□□
2. □□□ **Agent** □□□□□□□□**Coze / Dify / FastGPT**
   - □□□LLM □□□□□□□□□□□□
   - □□□□ Coze □□□ Agent□□□□□“□□□□□□□□□□□□”□
     - □□ A□LLM□□□□□□□□□□□ 5 □□□ Prompt□
     - □□ B□API□□□□ Runway / Seedance □□□□□□□□ API □□□□□
     - □□ C□API□□□□ ElevenLabs TTS □□□□□
     - □□ D□□□□□□□ FFmpeg □□□ CapCut API □□□□□
   - □□□□□□□□□□□□□□□□ Prompt □□□□□□□□“□□□□□”□□□□

3. □□□□□□**Make.com / n8n / Zapier**

    - □□□□□□□□□□□□□□□□
    - □□□□□ Notion □□□□□□□□□□□□□□□□□“Ready”□n8n □□□□□□ -> □□ Coze □□□□□ Agent -> □□□□ -> □□□□□ YouTube Shorts □ TikTok□

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7. □□□□□□

7.1 □□□□□

#	□□	□□□□□□	□□□□□□	□□□□□□
1	□□-□□□□□□	□□□□□	□ CLIP □□	⚠ □□□□
2	□□□□□	□□□	□ □□□□	□ □□□□
3	□□□□□□	AI □□□	⚠ □□□□	□ □ consistent ID
4	□□□□□□	AI □□□	□ □□□□	□ □□□□
5	□□□□□	□□/□□□	□ A/V sync	⚠ □□□□
6	□□□□□	AI □□□	⚠ VLM □□	□ □□□□
7	□□□□	□□□	□ NIQE/VMAF	⚠ □□□□
8	□□□□/□□	□□□	⚠ □□□□	□ □□□□□
9	**AI** □□□□	AI □□□	⚠ □□□□	□ □□□□□□
10	□□□□	□□□	⚠ VLM □□	□ □□□□

7.2 □□□□□□□

> *□ [□□□□□□□□□□□□□□□□□□□ **61**]*

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8. □□□□□□□

> *□ [□□□□□□□□□□□□□□□□□□□ **62**]*

> *"**AI** □□□□□□□□□□" — Reddit □□*
>
> *87% □□□□□□□□□□ Prompt □□□□□□ Prompt □□□□□□□□□□□□□□□□□ Prompt □□□□□□*

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9. □□□□□□ B □ A

9.1 □□□□

	□□ **A**□□□□□	□□ **B**□□□□□
□□	"□□□□□□□□□□□□□□"	"□ N □□□□□□□"
□□	□□→□□→□□→□prompt→□□	□□□□N□→□□→□□□

□□	□□□□□	MapReduce
□□□□	VISTA □□□□ 60%	□□"□□ 10 □ 1"
□□	□□□□ ~9 □□□	□□□□ ~1 □□□
□□□	□□□□□□□	□□□□□□□□

9.2 □□□□□Kling 3.0, $0.12/s, □□□□□ 60%, 6 □□/□□

□□	□□ A	□□ B (N=3)	□□ C (B+A)
□□/□	~$4.60	~$8.10	**~$6.00**
□□/□	~9 □□	~1 □□	**~2 □□**
□□□□	~100%	67.8%	**97%+**

9.3 □□□□□

> *□ [□□□□□□□□□□□□□□□□□□□□ 63](#)*

9.4 □□□□

> *□□□□□□□ **C□□ B □ A**□□□□□□□□□□*
>
> *□□□□□□□□□□□□B □ A □□□□□□□□□□□□□□□□□*
>
> *□□□□□□□Kling 3.0 □□□ API□*
>
> *□□□□□2026-03-14*

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10. □□□□□□□□

10.1 AI □□□□

□□	URL	□□□□	□□	API
Kling 3.0	klingai.com	4K/60fps+□□□+□□□□	Free 66 daily; Std $6.99/mo	□ ~$0.12/s
Seedance 2.0	dreamina.capcut.com	2K+12□□□□□+□□□□	Free 225 daily; Basic $18/mo	⚠ □□+□□□
Veo 3.1	aistudio.google.com	4K+□□□□+□□	AI Pro $19.99/mo	□ $0.15/s Fast
Runway Gen-4.5	runwayml.com	□□□□□□+60s□□□	Free 125 credits; Std $12/mo	□ $0.05/s Turbo
Sora 2	sora.com	□□□□(25s)+□□□+□□	Plus $20/mo	□ $0.10-0.50/s
Pika 2.5	pika.art	□□□□+□□□□	Free 80 credits; Std $8/mo	□ fal.ai

Hailuo	hailuoai.video	□□□□□	Std $9.99/mo	□ $0.19/video
Luma	lumalabs.ai	3D/VFX □□	Free 30 gens; Lite $9.99/mo	⚠ □□
Adobe Firefly	firefly.adobe.com	□□□□, Premiere □□	Creative Cloud	□
LTX Studio	ltx.studio	□□+□□□+□□□□	Free 800 credits; Lite $15/mo	□

10.2 □□□□

□□	URL	□□	□□
CapCut	capcut.com	AI □□	Free; Pro $19.99/mo
Descript	descript.com	□□□□□	Free; Pro $30/mo
OpusClip	opus.pro	□→□□□	Free; Starter $15/mo

10.3 AI □□ / □□

□□	URL	□□	□□
HeyGen	heygen.com	AI □□+175 □□	Free; Creator $29/mo
Synthesia	synthesia.io	□□□□□□	□□□□
ElevenLabs	elevenlabs.io	□□□□+□□	Free 10K; Pro $99/mo
edge-tts	github.com/rany2/edge-tts	□□ TTS	□□

10.4 □□□□

□□	GitHub	□□
Wan 2.2	Wan-Video/Wan2.2	VBench #1, 10GB □□, MIT
LTX-2	Lightricks/LTX-2	4K+□□, 12GB+, Apache 2.0
ComfyUI	comfyanonymous/ComfyUI	□□□□□□, GPL-3.0
CogVideoX	zai-org/CogVideo	□□, Apache 2.0
HunyuanVideo	Tencent/HunyuanVideo	□□ 13B
MoneyPrinterTurbo	harry0703/MoneyPrinterTurbo	55k stars, □□□□, MIT
NarratoAI	linyqh/NarratoAI	VLM □□□□+□□

10.5 □□□

□□	URL	□□

Pexels	pexels.com	□□□□, 200 req/hr
Pixabay	pixabay.com	□□□□

□□ A□□□□□

□□	□□
Kling AI + Kuaishou/Nasdaq	Kling 3.0 □□□□
Seedance 2.0 / Dreamina	□□□□□□□□
□□□□ + Atlas Cloud	Seedance API □□
Google AI Studio	Veo 3.1 □□/□□
Runway ML + releasebot.io	Gen-4.5 □□
Sora / OpenAI + PCMag	Sora 2 □□/□□
Pika	Pika 2.5 □□
NVIDIA CES/GDC 2026	LTX-2 + ComfyUI □□
ModelsLab + PiAPI	□□□ API □□
MoneyPrinterTurbo GitHub	□□□□□□
LTX-2 + TechPowerUp	□□□□□□
Wan 2.2 + Stackademic	□□□□□□
WaveSpeedAI	□□□□□□□□□
Reddit r/aivideo, r/StableDiffusion	□□□□□□/□□□□□
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08 — AI □□□□□□□□□□□□ (2026.03)

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Build□□□□	3-15 □	2-8 □	□□□□□□□□□□
Body□□□□	15-45 □	8-22 □	□□□□□□□
CTA□□□□	45-60 □	22-30 □	□□□□□□□□/□□/□□□

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- 30 □□□□80-100 □□□□□/ 80-100 words□□□□
- 60 □□□□150-200 □□□□□/ 120-150 words□□□□

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$0	Edge-TTS	`zh-CN-XiaoxiaoNeural` □□□□□□`zh-CN-YunxiNeural`□□□□□□	`en-US-JennyNeural`	`pip install edge-tts` → `edge-tts --text "□□" --voice zh-CN-XiaoxiaoNeural -o output.mp3`
$5.5/□	Fish Audio	□□□□□ + 50+ □□□□	□□□	fish.audio
$5/□	ElevenLabs	□□□□□□□	□□□□ **TTS**	elevenlabs.io
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Edge-TTS □□□□

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BGM□□□□□□	-10 □ -14	□□□□□□□□□
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□□	**URL**	□□
YouTube Audio Library	YouTube Studio □□	YouTube Shorts □□
Pixabay Music	pixabay.com/music	□□□□
Fesliyan Studios	fesliyanstudios.com	□□□□"□□□□□□"□□
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Uppbeat	uppbeat.io	□□□□□□

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Soundraw	soundraw.io	□□□□□/□□/□□
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YouTube Shorts	9:16 □□	1080×1920	□□
Instagram Reels	9:16 □□	1080×1920	□□
B □	16:9 □ 9:16	1920×1080 □ 1080×1920	16:9 □□□□
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TikTok	10:00-11:00 (EST) □□-□	19:00-21:00
YouTube Shorts	14:00-16:00 (EST)	20:00-22:00
Instagram Reels	11:00-13:00	19:00-21:00

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~$10/□	GPT-4o-mini	Fish Audio	CapCut □□	Kling □□□	Uppbeat	CapCut
~$50/□	Claude / GPT-4o	ElevenLabs	CapCut + □□ QA	Runway $28 / Kling $26	Envato Elements	CapCut / Premiere
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ElevenLabs	TTS □□
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IndexTTS-2	□□ TTS
CosyVoice	□□□□ TTS
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Kling AI	AI □□□□
Pexels	□□□□
Pixabay Music	□□□□
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- Artificial Analysis □□ Elo □□□□□□□□□□□
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1	**Runway Gen-4.5**	1247	Artificial Analysis □□
2	**Seedance 2.0** (□□)	1235	□□□□□□□□□□□□□□□□□□□□
3	Veo 3.1	1386	SiliconFlow Arena (□□□□□□□□□)
—	Kling 3.0	□□□□□□ #1	□□□□ #1□□□□□□

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- **Motion Brush**□□□□□□□□□□□□□□□□□□□□
- **Prompt**□□□ `[□□□□], [□□□□], [□□/□□], [□□/□□], [□□], [□□/□□], [□□□□]. [□□□□]`

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- **“□□□”□□□ 6** □□□ `[□□], [□□], [□□], [□□], [□□], [□□□□]` □
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□□□□□Veo 3.1 (□□□□□□□□□□□□□)

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- □□□□□□□□Prompt □□□□ `"with ambient café sounds, soft jazz music..."` □Veo □□□□□□□□□□□ □□□

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Runway	Unlimited	$76/□	□□□□□Gen-4/4.5 □□□
□□ **(Seedance)**	□□□	~□399/□	□□□□□□□□□□□□□□□□□
Veo 3.1	Google AI Ultra	$249.99/□	□□□□□4K □□

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- □□ TTS □□□□□□□□□
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Creator	$22/□	100K □□	□□□□
Pro	$99/□	500K □□	□□□□
Scale	$330/□	2M □□	□□□□

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- **CapCut Pro** / □□□□□□□□□□□ + □□ OK + □□□□□□
- **Premiere Pro + MOGRT** □□□□□□□□

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Artlist	artlist.io	~$16.60/□	□□□□□□□□□□□□□
Musicbed	musicbed.com	□□□□□	□□□□□□□□□□□□
Epidemic Sound	epidemicsound.com	$15/□	□□□□□□□□□□□□

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BGM□□□□□	**-24 □ -30 dBFS**	□□□□ 15-20 dB
BGM□□□□□□	**-12 □ -16 dBFS**	□□□□□□□□□□□
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□□□□	**CapCut Pro**	□□□□□□□□
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□□□	9:16 □ 3:4	1080×1920 □ 1080×1440	30fps	10-15 Mbps	H.264
□□□□□	9:16	1080×1920	30fps	10 Mbps	H.264
TikTok	9:16	1080×1920	30fps	12 Mbps	H.264
Instagram Reels	9:16	1080×1920	30fps	10 Mbps	H.264
YouTube Shorts	9:16	1080×1920	30fps	12 Mbps	H.264
□□□□	16:9	1920×1080	30fps	15-20 Mbps	H.264

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Runway / □□□□□	~$55-76	□□□□□□□□□□□□□□□□□□□□
ElevenLabs Pro	$99	□□□□□
MidJourney Standard	$30	□□□□□

Artlist	$17	□□□□□□
CapCut Pro	$20	□□+□□
Veo 3.1 □□	~$30-50	□□□□/□□□□□□
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4. **TTS &** □□□□□□□□ ElevenLabs API □□□□□□□ Python □□□□□□ FFmpeg □□□□□□□□

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- **Step 1**□□□□□□□□□□□□□ 0-5 □□□
- **Step 2**□□□□ 5 □□□□□□□□ **(Last Frame)** □□□□□□□□
- **Step 3**□□□□“□□”□□□□□□□□5-10 □□□**□□□□ (First Frame Prompt)**□□□□□□□□□□□□□□□“□□□□□□□□□□”□□
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3. □□□□□□ (Keyframe Interpolation)

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- **□□□□ (Fixed Seed)**□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Seed □□□□□□□□
- **□□□□□ (Storyboard)**□□□ Sora 2 □ Kling 3.0 □□□□□□□□□□□□□□“□□□□”□□

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□□□	Pexels + CLIP □□□□	Wan 2.2 □□□10GB GPU□	—
□□□□ **4K**	LTX-2□Apache 2.0, 12GB+□	Wan 2.2 5B□MIT, 10GB+□	—

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$0	DeepSeek	Pexels+CLIP	Edge-TTS	CapCut □□	Pixabay Music	CapCut
~$30	GPT-4o-mini	□□□□□	Fish Audio	CapCut	Uppbeat	CapCut
~$100	Claude	Runway Std $28	ElevenLabs $22	CapCut Pro $20	Artlist $17	CapCut Pro
~$300	Claude	Runway $76 + Veo	ElevenLabs $99	CapCut Pro	Artlist	Premiere
□□□	Ollama □□	Wan 2.2 / LTX-2	CosyVoice / IndexTTS-2	Whisper	□□□	FFmpeg

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2. □□□□□

□□□ A□□□□□□□15 □□/□□$0-1□

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□□□ B□AI □□□□□□□□□□□~$1-3/□□

2026.02 □□□□□□ — AI □□□□□□□□□+□□+□□□

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□□□ C□Agentic □□□□Coze/Dify □□□□

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□□□□□**Dify + SiliconFlow□□**

1. □□ SiliconCloud□□□ API Key
2. □ Dify □□□□□□ SiliconFlow □□
3. □□ Workflow□□□ SiliconFlow Video Generate □□
4. □□□□ `Wan-AI/Wan2.2-I2V-A14B` □□□ prompt □ image □□
5. □□ End □□□□□□ URL

□□□ D□ComfyUI □□□□□□□□/□□□□

□□□□□□□□□ API□□ 24GB+ GPU□

> *□ □□□□□□□□□□□□□□□□□□□ **82***

- NVIDIA GDC 2026 □□ ComfyUI App View + LTX-2.3 NVFP4 □□
- Wan 2.2 □ LTX-2 □□ ComfyUI □□□□□
- □□□□□□□□□□□□□□□□□

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3. □□□□

LLM □□

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□□□□□	□□ / **DeepSeek**	□□□□□□
□□□□□	**GPT-4o / Claude**	□□□□□
□□□	**DeepSeek**	□□□□
□□	**GPT-4o-mini**	□□□□

□□□□□Hook-Build-Body-CTA

□□	**60s**	**30s**	□□
Hook	0-3s	0-2s	□□□□□□ 3 □□□□□
Build	3-15s	2-8s	□□□
Body	15-45s	8-22s	□□□□□□
CTA	45-60s	22-30s	□□□□

6 □ Hook □□

□□	□□	□□
□□□	"99% □□□□□..."	□□□□
□□	"□□□□□□..."	□□
□□	"□□□ X □□□"	□□□□
□□□□	"□□□□□□□□——"	□□/□□
□□□	"□□□□□□□□□□□"	□□□□

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Prompt □□

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4. AI □□□□ + Prompt □□

□□□□□Image-to-Video > Text-to-Video

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1. □ MidJourney/DALL-E □□□□□□□□□□
2. □□□ Runway/Seedance/Veo □ Image-to-Video
3. □□□□ ~~60-73% □□□ **~~85-90%**

□□□□ Prompt □□

Runway Gen-4.5

> *□ □□□□□□□□□□□□□□□□□□□ 84*

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- □ □□□ negative prompt□□□□□□□□□□
- □ □□□□□□□□□□□□□□□□
- □ Image-to-Video □□□□□□□□□□□□
- □ □□ prompt □□□□□□□□□□□□□□

Seedance 2.0□□□□

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□□□ **—** @ □□□□□

- `@□□1 □□□□` — □□□□
- `@□□2 □□□□□□` — □□□□□
- `@□□1 □□□□□□` — □□□□□□
- □□ 12 □□□□□

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- □□□□□□□ 30-40%□□□□□□□□□
- □□□□□ **5-8** □□10 □□□□□ — "10 □□□□□"□
- □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

Veo 3.1

> *□ □□□□□□□□□□□□□□□□□□□ 86*

Veo □□□□□

- □□□□□□□□prompt □□□□□□□□□
- □□□□□□□□□ + □□ + □□
- □ □□□ "No subtitles, no text overlay"□□□□□□□□□
- □□□□□□□□□□prompt □□□□□□□□□

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□□	Runway	Seedance	Veo
□□	□□	□□□□	□□
Negative prompt	□	□□□	"No..." □□
□□□□	□	□	□
□□	60s (Gen-4)	5-12s	~8s
□□□□	□□□□□□□	□□@□□□	□
Prompt □□□	□□□□□□	□□+□□□□	□□□□+□□□□

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5. □□□TTS□

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Edge-TTS	7/10	□ □□/□□	□ Jenny	$0	□
Fish Audio	9/10	□ 50+□□□□	□	$5.5/□	□
ElevenLabs	9.5/10	⚠ □□	□ □□	$22-330/□	□ 30s □□
IndexTTS-2	9/10	□ □□□□□	—	□□□□□	□
CosyVoice	9/10	□ 18 □□	□	□□□□□	□ □□□

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□□□□	□□	□□	□□
□□□□	1.0-1.1x	□□	□□□□ 0.3-0.5s
□□/□□□	1.15-1.25x	□□ +5%	□□□
□□□□	0.95-1.05x	□□□□□□	□□□ 0.5-1.0s

Edge-TTS □□□□

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⚠ Edge-TTS □□□□□□□□□ API□□□□ **SLA**□□□□□□□□□□□□

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6. □□

2026 □□□□□□□ OK □□

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□□	□□□□□□□□□□□/Montserrat□
□□	□□ 8-12 □□□□ 3-5 □
□□	□□ 2 □
□□	□□□□ 1/3□□□□□ UI□
□□□	□□□□□□□□□□□□
□□	□□□□□ **0.1-0.2s** □□
□□	□□□□□□□ OK □□

CapCut □□□□

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7. □□□□□

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□□□	□□ **(dBFS)**
□□/□□	**-10 □ -12**
BGM□□□□□□	**-24 □ -30**□□□□□ 15-20dB□
BGM□□□□□□	**-12 □ -16**
□□	**-14 □ -20**
□□□□	**-14 LUFS**

□□□□□□

□□	URL	□□
YouTube Audio Library	YouTube Studio □□	YouTube Shorts □□
Pixabay Music	pixabay.com/music	□□□□
□□□	aigei.com	□□□□□□□□+BGM
Fesliyan Studios	fesliyanstudios.com	□"□□□□□□"□□
Uppbeat	uppbeat.io	□□□□□□

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□□	□□	□□
Artlist	~$17/□	□□□□□□□□□□□□
Epidemic Sound	~$15/□	□□□□□□

Musicbed	□□□	□□□□

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□□	□□	□□
Whoosh	□□□□	Pixabay / □□□
Riser	□□□□	Pixabay / □□□
Hit/Impact	□□□□	Pixabay / □□□
Pop/Ding	□□□□	Pixabay / □□□

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8. □□□□

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□□	□□	□□□
□□/TikTok/Shorts/Reels	9:16 □□	1080×1920
B □	16:9 □ 9:16	1920×1080
YouTube □□□	16:9 □□	1920×1080

□□□□□□□□□□□□ **9:16** □□□□□□□□□□□□□□

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□□□/□□	0.5-2s
□□/□□	2-5s
□□□□□□	3-6s

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□□	□□□□
□□□	80%+
□□□□	10%
□□/□□	5%
□□□□	5%

2026 □□□□□□□□□□□□□□□□□□□□□□

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9. □□□□

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2	□□-□□□□	□□□□□□□□□□□□
3	□□□□	□□□□□□□□□
4	□□□□	□□□□□BGM □□
5	□□	□□□□□/□□/□□
6	□□□□	□□□□□□□□□□□□□□

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VMAF (Netflix)	□□□□ vs □□	`ffmpeg`□□□ libvmaf□	□□□
DNSMOS (Microsoft)	□□/□□□□	`pip install torchmetrics[audio]`	□□□
SyncNet	□□□□□□	`pip install syncnet-python`	□□□
VBench	AI □□ 16 □□□□	`pip install vbench`	□□□

VMAF □□□□

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□□	□□
95+	□□□□□□□□
84-92	□□UGC □□□
70-84	□□□□□□□
<70	□□□□□□□

DNSMOS □□□□

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A/V □□□□

□□	□□
±20ms	□□□□
±45ms	□□□□□□□□□□

±80ms+	□□□□

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10. □□□□□□□□

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□□	□□□□	□□
□□	18:00-22:00	7:30-8:30, 12:00-13:00
TikTok	10-11 AM EST □□-□	7-9 PM
YouTube Shorts	2-4 PM EST	8-10 PM

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□□	1080×1920	30fps	12-15 Mbps	H.264
□□□	1080×1920 □ 1080×1440	30fps	10-15 Mbps	H.264
TikTok	1080×1920	30fps	12 Mbps	H.264
YouTube Shorts	1080×1920	30fps	12 Mbps	H.264

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- □□□ **SEO**□□□□□□□□□□□□□ → □□□□□□□

A/B □□□□

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□□□	□□□	□□□□
1	Hook□□ 3 □□	3 □□□□ > 30%
2	□□□□□15s/30s/60s□	□□□ > 40%
3	□□/□□□	□□□
4	CTA □□□□□	□□□

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- □□□□□□ 1000+ □□□□□□
- TikTok □ 48-72 □□□□□YouTube Shorts □ 1-2 □

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3 □□□□	Hook □□□	> 30%
□□□□□□	□□□□□	> 50% □□□□
□□□	□□□□□□	> 40% (30s □□)
□□□	□□+□□+□□/□□	> 5% (TikTok)
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□□□□□□	□□□□□□	□□
Virlo	AI □□□□ + □□□□	□□
Exolyt	TikTok □□□□	□□
Brand24	□□□□ + □□□□	□□

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11. □□□□□□□□□

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Coze□□□□□□□□□□

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Dify + SiliconFlow □□□□

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n8n □□□□□

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Pixelle-Video□□□□□□□□□□□□ AIDC-AI□

GitHub: AIDC-AI/Pixelle-Video | □□□: Apache 2.0

□□□□ → □□□□ → AI □□/□□ → TTS □□ → BGM → □□□□□

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□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ TTS□image-to-video□

MCP Server□□ Claude/Cursor □□□□□□

MCP Server	□□	□□

Kling AI MCP	13+ □□□□□□□□□□□□□□□□□□□□□□	GitHub
Remotion MCP	React □□□□□□□□3D/□□/□□	remotion.dev
Remotion Video Gen	Claude □□□□□□□□□□	MCP Market

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ComfyUI Wan 2.2 □□□□

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□□□□□□□□ ComfyUI-GGUF, ComfyUI-KJNodes, ComfyUI-VideoHelperSuite, ComfyUI-mxToolkit, ComfyUI-Frame-Interpolation, ComfyUI-wanBlockswap, ComfyUI-MagCache

□□□ Steps 15-30, Guidance Scale 6, □□□ 720p

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□□	□□	□□
□□AI	□□□□□□□□□	20 □/□□□□□□□□
□□	□□□□□□+□□	□□"□□ 3 □"□□
□□	□□□□□□□	CapCut □□□□
Pixelle-Video	□□□□□□□□	Apache 2.0□□□□

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12. □□□ ROI □□

□□□□□□□□□30s□6 □□□

□□	□□□□	□□	□□	□□	□□
Pexels+Edge-TTS	$0	$0	$0	$0	**$0**
□□□□□	$0	$0	$0	$0	**$0**
Runway Std	~$3.00	$0.01	$0	$0	**~$3**
Runway+□□□□(N=2)	~$6.00	$0.01	$0	$0	**~$6**
□□□□	$500-2000	$50-200	$20-50	$30-100	**$600-2350**

□□ ROI□□□□□□□□□□

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13. □□□□□□□□

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□ prompt □□□□	AI □□□□□□□□□□□□□□□□□□□□□□
□ Kling "4K/60fps" □□	Reddit □□□□□□□□□ 1080p□□□□□□□
□□□ BGM □□□□□	□□□□□□□□
Seedance □□□□ 10 □	"10 □□□□□□"——10 □□□□/□□□□□
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Image-to-Video > Text-to-Video	□□□□□□□□□□□ 60% □□□ 85%+
5-8 □□□□□	□□ 10 □□□□□□□□□□□□□□□□

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Artificial Analysis	□□□□ Elo	★★★★★
Video Arena	□□□□□□	★★★★☆
Reddit r/aivideo	□□□□□□	★★★★☆□□□□□□□□□□□□□□
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GitHub Stars/Issues	□□□□□□□□	★★★★★
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36kr/TechCrunch	□□□□	★★★☆☆

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11 — □□□□□□□□□□□□□□□□□□□□□ (2026.03.14)

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□□□ MoneyPrinterTurbo□Pixelle-Video□n8n □□□□ Luma Agents□□□□□□□□□□

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□□□□	□□□□□□□□□	ComfyUI □□□	**Agent** □□□□□Luma Agents / Coze Agent □□□□□□+□□□□□
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1	**Pixelle-Video**	3.1k	2026-02-04 (push 03-08)	**AI □□**（FLUX+Wan 2.1）	Apache 2.0	□□□□□□□
2	**MoneyPrinterTurbo**	50.2k	2025-12-14 ⚠	Pexels □□□	MIT	□□□□□□□□□□
3	**NarratoAI**	8.3k	**2026-03-10** □	□□□□□□	□□□	□□□□/□□
4	**KrillinAI**	9.7k	2026-02-08	□□□□□	GPL-3.0	□□□□/□□
5	**Wan 2.1**	15.6k	**2026-03-05** □	AI □□	Apache 2.0	□□□□□□□□□
6	**LTX-Video**	9.6k	2026-01-05	AI □□	Apache 2.0	□□□□□□□+4K
7	**SkyReels-V2**	6.6k	2026-01-29	AI □□	Custom	□□□□□□□
8	**HunyuanVideo**	11.8k	2025-11-21	AI □□	Custom	□□□□□

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- **MoneyPrinterTurbo** □□□□□□3 □□□□□□□□□□□□□□□ Release v1.2.6
- **Pixelle-Video** □□□□□□□□□□□ — AI □□□□□□□□□□□+ Apache 2.0 + Windows □□□□□
- **NarratoAI** □□□□□□□□□□/□□□□□□□□□□□□□ Release v0.7.6
- **Wan 2.1** □□□□□□□ — VBench □□□ □□ Sora□86.22% vs 84.28%□□1.3B □□ □□ 8GB □□

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- **Image-to-Video** □□□□ **Text-to-Video** — □□□□□□□□□□□□□□□□ ~60% □□ ~85%+□□□□Magic Hour I2V vs T2V □□
- **Kling "Leap-Frog"** □□□□□ — □□□□□□ → Topaz Video AI □□ → □□□□□ I2V □□□ → □□□□□□□□□Reddit r/aivideo □□
- **Veo** "□□□□" — prompt □ "no background music"□□□□□□□□+□□□□□□□□□□□□□□GlobalGPT Veo 3.1 □□□□□□
- □□□□□ **5-8** □ — □□ 10 □□□/□□□□□□"10 □□□□□"□□□□□Seedance 2.0 □□□□□

□□ 2□"The Factory"□□□□□□□□

> ***n8n** / **Make.com** □□□□□□□□□□ **10+**□□ □□□n8n.io □□□□Medium n8n □□□□*

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□□	n8n Schedule	n8n.io	□□/□□□□
□□	OpenAI/Claude □□	n8n □□	JSON □□
□□	PiAPI (Flux 2)	piapi.ai	□□□□
□□	PiAPI (Kling)	piapi.ai	I2V □□
□□	ElevenLabs □□	elevenlabs.io	TTS
□□	Creatomate	creatomate.com	□□□□
□□	Blotato □□□□	n8n Settings → Community Nodes → `n8n-nodes-blotato`	□□□

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1. `npx n8n` □ Docker □□ n8n
2. □□□□□□□ `n8n-nodes-blotato`
3. □□ Creatomate → □□□□□□ → □□ Template ID
4. □□ n8n □□ #3121□AI □□□□□□□□□
5. □□□□ API Key → □□ Google Sheet → □□

□□ 3□"The Aggregator"□□□□□□□

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Higgsfield AI	higgsfield.ai	15+	□□□	Cinema Studio□□□□□□□Soul ID□□□□□□□□
Luma Agents	lumalabs.ai	□□□	$30-300/□	**2026-03-05** □□□□ Agent □□□□□□□□□□
Abacus.AI ChatLLM	abacus.ai	20+	$10/□	□ prompt □□□□□□RouteLLM □□□□□

Higgsfield AI □□□□□

- □□ 15+ □□□Sora 2, Kling 2.6, Veo 3.1, FLUX.2 □□
- **Cinema Studio 2.0**□□□□□□□□□□□ARRI, RED, Sony□□□□□□□35mm, 50mm, 85mm□
- **Character ID / Soul ID**□□□□□□□□□□□□□+□□+□□+□□
- **Shots** □□□□□ prompt □□□□□□□□□

Luma Agents — □□□□□□□2026-03-05 □□□□

- Agent □□□□□□Ray3.14, Veo 3, Sora 2, Kling 2.6 □□□□□□BusinessWire □□□MediaPost □□
- □□ prompt → Agent □□□□□ + □□ + □□

- □□□"AI □□□□"

□□ 4□"The Local Studio"□□□□□□□

> ***ComfyUI** + □□□□□□□□□□□ API□□ RTX 4090□*
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> *□ □□□□□□□□□□□□□□□□□□□ 104*

□□ **ComfyUI □□□**

□□	GitHub	Stars	□□
WanVideoWrapper	kijai/ComfyUI-WanVideoWrapper	**6,168**	Wan 2.1 T2V/I2V, 1025 □□□, SteadyDancer, LoRA
LTXVideo	Lightricks/ComfyUI-LTXVideo	3,300	LTX-2 □□□□□□
Kie-API	gateway/ComfyUI-Kie-API	21	Kling 3.0 Motion Control I2V
VideoHelperSuite	Kosinkadink/ComfyUI-VideoHelperSuite	—	□□□□/□□□□
Frame-Interpolation	Fannovel16/ComfyUI-Frame-Interpolation	—	□□/□□□

2026.03 ComfyUI □□□□□□□□□NVIDIA GDC 2026□□

- NVIDIA RTX Video Super Resolution □□ — □□ 4K □□
- NVFP4/FP8 □□ — 2.5x □□□□□60% □□□□
- App View — □□□□□□□□□□□□□□
- LTX-2.3 □□□□□□

□□□□□□**Pixelle-Video**

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1. Kling "Leap-Frog" □□□□□

□□□Reddit r/aivideo□Kling AI □□□□

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2. Veo 3.1 "□□□□"

□□□GlobalGPT Veo 3.1 □□□Superprompt Veo 3 Prompt Guide

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□□□□□□□□□□**Veo 3.1□□**

- □□□□□ `A woman says, "Welcome to the future."` — □□ + □□□□□

- □□□□□□□□□□□
- Fast □□ ($0.20/clip) □□□□□□□ — □ Standard □□□□

3. Runway Motion Sketch

□□□Curious Refuge Runway Gen 4.5 □□□□Runway □□□□

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□□ Runway □□ **Story Panel Upscaler** — □□□□□□□□□□□□□□□□□□□□□□□Runway Workflows Tab □□

4. Higgsfield "Soul ID"

□□□Higgsfield AI□Gaga Art - Higgsfield □□

□□□□□□□□Character ID□□□□□□□——□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ "character consistency" □□□□□□□□□□□

5. Seedance □□□□□□□

□□□Seedance 2.0 □□□□□□□□□ AI □□□Binance Seedance □□

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- `@□□2 □□□□□□` — □□□□□□
- `@□□1 □□□□□□` — □□□□□□
- □□ 12 □□□□□

6. □□□□□□□□□□□□□□□

□□□Reddit r/StableDiffusion□□□ YouTube AI □□□□□□

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□□□□□□□□□□□□□□□□□□□□□ Kling □ "Leap-Frog"□□ Seedance □ "□□□□□"□□ Wan 2.1 ComfyUI □□□ first-last-frame-to-video □□□

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□□□□□□□	Pixelle-Video + Ollama	GitHub / Ollama	□□□□Apache 2.0
□□□□□□□□	MoneyPrinterTurbo	GitHub	50k stars□□□□□
□□□□/□□	NarratoAI	GitHub	□□□□2026.03.10□□VLM □□□□
□□□□/□□	KrillinAI	GitHub / klic.studio	100+ □□□□□□□
□□□□□	n8n + PiAPI + Creatomate	n8n □□ / PiAPI / Creatomate	□□ 10+ □□□□□□

□□□□□□□□	Higgsfield + Runway + Veo	Higgsfield / Runway / Veo	□□ 15+ □□
Agent □□□□	Luma Agents	lumalabs.ai	2026.03.05 □□□□Agent □□□□
□□□□□ **4K**	ComfyUI + Wan 2.1 + LTX-2	ComfyUI / WanWrapper / LTX	RTX 4090 □□□□
□□□□□□	Coze + □□ + CapCut	Coze / □□ / CapCut	□□□□□□
□□□□□□	SkyReels-V2	GitHub	□□□□□□□□□□□□
□□□ **Kling API**	Kie.ai	kie.ai	$0.025/s□□□□□□ ~60%

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Wan 2.1	GitHub	2026-03-05 □	□□	□□□□□□□□
WanVideoWrapper	GitHub	2026-03-14 □	□□□□	ComfyUI □□□□
NarratoAI	GitHub	2026-03-10 □	□□□□	□□□□□
ComfyUI-Kie-API	GitHub	2026-03-13 □	□□	Kling API □□
KrillinAI	GitHub	2026-02-08 □	□□	□□□□□
Pixelle-Video	GitHub	2026-02-04 □	□□	□□□□□□
SkyReels-V2	GitHub	2026-01-29 □	□□	□□□□□
LTX-Video	GitHub	2026-01-05 □	□□	□□□
MoneyPrinterTurbo	GitHub	2025-12-14 ⚠	□□	□□□□□□□□
HunyuanVideo	GitHub	2025-11-21 ⚠	□□	□□□□□□□

□□□□□□□□□□2026-03-14 □□□□ Grok □□□

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Luma Agents	□□□□	—	**2026-03-05** □□□□**9** □□□□□□□□□□ Agent□□□ Sora 2/Veo 3.1/Kling/ElevenLabs□□□□□□□□□□$30-300/□□□□□ BusinessWire□MediaPost
ComfyUI-WanVideoWrapper	□□□□	**6,168**	kijai □□□140 □□□□□□ 1025 □□□□□□ 81 □+16 □□□□ SteadyDancer □□□□LoRA□VRAM □□
ComfyUI-Kie-API	□□□□	21	Kling 3.0 Motion-Control I2V □□□2026-03-13 □□□□□□□□ □□□□□Kie.ai □□□□
Grok Imagine API	□□□□	—	xAI □□/□□ API□$0.70/10s □□□Aurora □□□□□ A/V □□□⚠ Grok □□□□□□□□□□□□□□□□□xAI □□□DataCamp
Abacus.AI ChatLLM	□□□□	—	$10/□ 20K credits□□□□□□ □GPT/Gemini/Claude/Veo/Runway □□□□□□Abacus.AI □□
Kie.ai	□□□□	—	Kling API □□□$0.025/s □□Kling 2.1□□□□□□□□ 60%□□□□ Kie.ai □□□
n8n □□ #3121	□□□□	—	"AI □□□□□□"□Google Sheet → OpenAI → Flux → Kling I2V → ElevenLabs → □□

□□□□□Grok □□□□□

Grok □□	□□□□	□□□□
"Wan 2.6 □□□□"	□ **Wan 2.6** □□□ **API**□□□□□□□□□□□□□□□□ □ Wan 2.1 (15.6k □) □ Wan 2.2 (14.6k □)	GitHub Wan-Video □ □
"□□□□ MoneyPrinterTurbo/Pixelle-Video"	□ □□□MPT 50K stars □□□□Pixelle-Video □□□□	GitHub stars/commit □□
"□□ 500+ □□"	⚠ □□□□□□□□□□□□□□	□□□□□□□
"1025 □□□"	⚠ □□□□ 1025 □□□□□□ 81 □□□□□ 1025 □□□	WanVideoWrapper □ □

2026.03 □□□□□□□□□□□□□□

> *□ □□□□□□□□□□□□□□□□□□ **110***

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□□	□□ n8n □□ + Kie.ai Key	n8n □□ #3121 + Kie.ai	**1** □□
□□	ComfyUI + Kie-API + WanWrapper	Kie-API + WanWrapper	□□
□□□	n8n + ComfyUI + Luma Agents	n8n + ComfyUI + Luma	1-2 □
□□□	Pixelle-Video + Ollama	Pixelle-Video + Ollama	2-3 □□

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□□	□□	□□
GitHub API □□□□	□□□ GitHub □□	□□□□
Curious Refuge□AI □□□□□□□□	curiousrefuge.com	YouTube □□
AIFire□AI □□□□□□□□	aifire.co	□□□□
VO3.AI□AI □□□□□□□□	vo3ai.com	YouTube □□
SoulAI Writes□AI □□□□□	soulaiwrites.com	□□□□
Reddit r/aivideo	reddit.com/r/aivideo	□□□□□□
Reddit r/StableDiffusion	reddit.com/r/StableDiffusion	□□□□
n8n □□□	n8n.io/workflows	□□□□□
Createmate + n8n □□	createmate.com	□□□□ API
Higgsfield AI	higgsfield.ai	□□□□
Luma Labs	lumalabs.ai	Agent □□
ComfyUI □□	comfy.org	□□□□
NVIDIA GDC 2026	nvidia.com	□□□□
Bilibili AI □□□□	bilibili.com	□□□□
Superprompt□Veo 3 Prompt□	superprompt.com	Prompt □□
GlobalGPT□Veo □□□□□□□□	glbgpt.com	□□
Seedance □□□□□	chat4o.ai	□□
Magic Hour (I2V vs T2V)	magichour.ai	□□□□
Topaz Labs□□□□□□	topazlabs.com	□□□□
Artificial Analysis□□□□□□	artificialanalysis.ai	□□□□

xAI Grok Imagine □□	docs.x.ai	API □□
Kie.ai□Kling API □□□	kie.ai	API □□
PiAPI□Flux/Kling API□	piapi.ai	API □□
Abacus.AI	abacus.ai	□□□□□

2026 AI □□□□□□□□□□□ (The Ultimate AI Video Handbook)

> *□□□2026.03 □□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□“□□□□”□□□□□□□□□□□□□□□□□□□

1. MoneyPrinterTurbo (MPT) □□□□□□□

- □□□MPT□55k+ Stars□□“□□□□□□□□”□□□□□□□□□“□□□□□□□”□□□□□□
- □□□MPT □□□□ Pexels □□□□□□□□□□□□□□□□□□□□□□□“□□□□”□□□□□□□□□□□□□ AI □□□□□□□□□□ 2026 □□□□□□□□□□□□□□□□□□□□□

2. □□ MPT□□□□□□□□□□□□□ Repo□

□□□□□□□□“Agentic Workflow□□□□□□□□□”□□□ GitHub □□□□□□□□□□□□□□

- **[Short Video Maker (gyoridavid) / MCP □□]**
  - □□□□□□□□□□□□□ MPT□
  - □□□□□□□□□□□□□□□ (MCP)□□□□ Claude □ n8n □□□□ API □□□□□□□□ Kokoro TTS □ Remotion □□□□□□□□
- **[AI-Faceless-Video-Generator (SamurAIGPT)]**
  - □□□□□□□/□□□□□□□□□□□□
  - □□□□□ SadTalker□□□□□□□□□□□ → □□ → □□□□□□□□□□“□□□□”□□□□□□□□□□□□□□□□□□□□□□□□□□□□
- **[Wan 2.6 + n8n □□□]**
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  - □□□□□□□□□ Wan □□□10GB □□□□ 720p□□□□ n8n □□ `GPT-4o □□ → □□ Wan GPU □□□□□ → FFmpeg □□ □□□□□□□ Runway □□□□□□□□□□□□□
- **ComfyUI (□□□□□)**
  - □□□□□□□□□□□□□□□□□□
  - □□□□ LTX-2 □ HunyuanVideo □□□□□□□□□□□□“Midjourney □□ → □□□□ → □□□□ → Topaz □□”□□□□□□□□□□□□□□□□□□

□□□□MoneyPrinterTurbo □□□ —— □□ Seedance □□□□ SOP

□□ MPT □□□ Pexels □□□□□□□□□□□□□□□□□“□□ **Seedance 2.0** □□□□□□□□□□”□□□□□□□□

□□□□ MPT □□□□□□□□□□□□□ **Coze/Dify + □□ API□□ PiAPI□** □□□□□□□□□□□□□□□□□□□□□□□□ (SOP)□

SOP□□□“Seedance + □□□”□□□□□□□□

1. □□□□□
   - □□ Coze □ Dify□□□□□□□“□□□□□□□□□”□ Agent□
2. □□ **LLM** □□□□□□□□□□□□□□
   - `□□□ {□□□□}`
   - □□□□□□□□ LLM □□ 5 □□□□□ **Seedance 6**□□□□□ `[□□], [□□], [□□], [□□], [□□], [□□□□]` □
3. □□□□□□□□□□**Seedance API□□**
   - □□□□□□□□ Seedance API□□□□□□□□ API□□
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   - *(□□□□□□□□□□□□□□□□□□□

6. □□□□□Agent □□□□□

4. **[15-20s □□]** □□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□

□□□□□□□□□□ElevenLabs□

- □□□□□□□□□□□□□□□□□□□□□□□□□□ 0.9x□□□□□□“□□□□□□□□□□□□□□”□
- □□□□□□□□□ `Bass Drop` □□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□CapCut / PR□

□□□□□□□□□□□□□□□“□□□□□□”□□□□□□□□□□□□□□□□□□□

```
    v
[LLM Script Generation] -- GPT-4 / Claude / DeepSeek / local LLM
    |
    v
[Asset Sourcing] -- Pexels / Pixabay / Storyblocks API
    |
    v
[TTS Narration] -- Edge-TTS / ElevenLabs / Azure
    |
    v
[Subtitle Generation] -- Whisper / stable-ts
    |
    v
[Composition] -- FFmpeg + MoviePy + OpenCV
    |
    v
[Output] -- MP4 (1080x1920 vertical)
```

□□□□ 4 (□□: `02-evaluation-tools.md`)

```python
pip install vbench
# or from source:
git clone https://github.com/Vchitect/VBench.git
cd VBench && pip install -e .
```

□□□□ 5 (□□: `02-evaluation-tools.md`)

```python
from vbench import VBench

evaluator = VBench(device="cuda")

# Evaluate a single video against its prompt
results = evaluator.evaluate(
    video_path="output.mp4",
    prompt="A golden retriever running on a beach at sunset",
    dimensions=["subject_consistency", "temporal_flickering",
                "motion_smoothness", "aesthetic_quality"]
)

for dim, score in results.items():
    print(f"{dim}: {score:.4f}")

# Output example:
# subject_consistency: 0.9234
# temporal_flickering: 0.8871
# motion_smoothness: 0.9012
# aesthetic_quality: 0.7845
```

□□□□ 6 (□□: `02-evaluation-tools.md`)

```
git clone https://github.com/VQAssessment/DOVER.git
cd DOVER && pip install -e .
# Download pretrained weights
python download_pretrained.py
```

□□□□ 7 (□□: `02-evaluation-tools.md`)

```
import dover

model = dover.DOVER(pretrained=True, device="cuda")

scores = model.evaluate("output.mp4")
print(f"Technical quality: {scores['technical']:.4f}")  # 0-1
print(f"Aesthetic quality: {scores['aesthetic']:.4f}")  # 0-1
print(f"Overall (DOVER):   {scores['overall']:.4f}")    # 0-1
```

□□□□ 8 (□□: `02-evaluation-tools.md`)

```
pip install open_clip_torch
```

□□□□ 9 (□□: `02-evaluation-tools.md`)

```
import open_clip
import torch
from PIL import Image
import cv2
import numpy as np

# Load model
model, _, preprocess = open_clip.create_model_and_transforms(
    "ViT-H-14", pretrained="laion2b_s32b_b79k"
)
tokenizer = open_clip.get_tokenizer("ViT-H-14")
model.eval()

# Extract frames from video
def extract_frames(video_path, n_frames=16):
    cap = cv2.VideoCapture(video_path)
    total = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    indices = np.linspace(0, total - 1, n_frames, dtype=int)
    frames = []
    for idx in indices:
        cap.set(cv2.CAP_PROP_POS_FRAMES, idx)
        ret, frame = cap.read()
```

```
        if ret:
            frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
            frames.append(Image.fromarray(frame))
    cap.release()
    return frames

# Compute similarity
frames = extract_frames("output.mp4", n_frames=16)
text = tokenizer(["A golden retriever running on a beach at sunset"])

with torch.no_grad():
    frame_features = torch.stack([
        model.encode_image(preprocess(f).unsqueeze(0))
        for f in frames
    ]).squeeze(1)
    text_features = model.encode_text(text)

    # Normalize
    frame_features /= frame_features.norm(dim=-1, keepdim=True)
    text_features /= text_features.norm(dim=-1, keepdim=True)

    # Average similarity across frames
    similarities = (frame_features @ text_features.T).squeeze()
    avg_similarity = similarities.mean().item()

print(f"Text-video alignment: {avg_similarity:.4f}")
# Typical range: 0.20-0.35 for good alignment
```

□□□□ 10 (□□: `02-evaluation-tools.md`)

```
git clone https://github.com/VQAssessment/COVER.git
cd COVER && pip install -e .
```

□□□□ 11 (□□: `02-evaluation-tools.md`)

```
pip install scenedetect[opencv]
```

□□□□ 12 (□□: `02-evaluation-tools.md`)

```
from scenedetect import detect, ContentDetector

# Detect scene changes
scene_list = detect("output.mp4", ContentDetector(threshold=27.0))

print(f"Number of scenes: {len(scene_list)}")
for i, scene in enumerate(scene_list):
    print(f"  Scene {i+1}: {scene[0].get_timecode()} - {scene[1].get_timecode()}")
```

```
# For quality checking: unexpected scene changes in a continuous shot
# indicate generation artifacts
if len(scene_list) > expected_scenes:
    print("WARNING: Unexpected scene breaks detected")
```

□□□□ 13 (□□: `02-evaluation-tools.md`)

```
pip install stable-ts
```

□□□□ 14 (□□: `02-evaluation-tools.md`)

```
import stable_whisper

model = stable_whisper.load_model("large-v3")
result = model.transcribe("audio.wav")

# Get word-level timestamps
for segment in result.segments:
    for word in segment.words:
        print(f"[{word.start:.2f} - {word.end:.2f}] {word.word}")

# Validate against existing subtitles
result.to_srt_vtt("validated_subtitles.srt")
```

□□□□ 15 (□□: `02-evaluation-tools.md`)

```
pip install imagehash
```

□□□□ 16 (□□: `02-evaluation-tools.md`)

```
import imagehash
from PIL import Image
import cv2

def check_frame_consistency(video_path, threshold=15):
    """Detect frames with abnormal visual changes."""
    cap = cv2.VideoCapture(video_path)
    prev_hash = None
    anomalies = []
    frame_idx = 0

    while True:
        ret, frame = cap.read()
        if not ret:
```

```
            break
        img = Image.fromarray(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
        curr_hash = imagehash.phash(img)

        if prev_hash is not None:
            diff = curr_hash - prev_hash
            if diff > threshold:
                anomalies.append((frame_idx, diff))

        prev_hash = curr_hash
        frame_idx += 1

    cap.release()
    return anomalies

anomalies = check_frame_consistency("output.mp4")
if anomalies:
    print(f"Found {len(anomalies)} frame consistency anomalies")
```

□□□□ 17 (□□: `02-evaluation-tools.md`)

```
pip install pyiqa
```

□□□□ 18 (□□: `02-evaluation-tools.md`)

```
import pyiqa

# Available metrics include: musiq, clipiqa, niqe, brisque, etc.
metric = pyiqa.create_metric("musiq", device="cuda")

# Score individual frames
score = metric("frame_001.png")
print(f"Frame quality (MUSIQ): {score.item():.4f}")
```

□□□□ 19 (□□: `02-evaluation-tools.md`)

```
Quality Gate Score = w1 * DOVER_overall
                   + w2 * OpenCLIP_similarity
                   + w3 * VBench_subset_avg
                   + w4 * (1 - flickering_anomaly_rate)
                   + w5 * subtitle_sync_accuracy
```

□□□□ 20 (□□: `02-evaluation-tools.md`)

```
Layer 1: Programmatic checks (ffprobe, imagehash, pysrt)
    |
```

```
    PASS? --yes--> Layer 2: ML model checks (DOVER, OpenCLIP, VBench subset)
    |                  |
    FAIL               PASS? --yes--> Output
    |                  |
    v                  FAIL/UNCERTAIN
  Retry                    |
                           v
                       Layer 3: VLM review (Gemini / GPT-4V)
                           |
                        PASS? --yes--> Output
                           |
                           FAIL --> Correction Agent --> Retry
```

□□□□ 21 (□□: `03-self-correction-patterns.md`)

```
                   +------------------+
                   |  Generation      |
                   |  (Veo 3)         |
                   +--------+---------+
                            |
            +--------------v--------------+
            |                             |
    +---------v---------+   +---------------v-----------+
    | Visual Fidelity   |   | Audio Fidelity            |
    | Reviewer          |   | Reviewer                  |
    +---------+---------+   +---------------+-----------+
            |                             |
            +---------+   +--------------+
                      |   |
                +-----v---v-----+
                | Context       |
                | Preservation  |
                | Reviewer      |
                +------+--------+
                       |
                +------v--------+
                | Reasoning     |
                | Agent         |
                | (Gemini 2.5)  |
                +------+--------+
                       |
                  +----v----+
                  | Re-gen  |  (if needed)
                  +---------+
```

□□□□ 22 (□□: `03-self-correction-patterns.md`)

```
Input
  |
  v
```

```
[Generator] --> Output
                  |
                  v
              [Evaluator]
                  |
            +-----+-----+
            |           |
          PASS        FAIL
            |           |
            v           v
        Output    [Structured Feedback]
                        |
                        v
                  [Optimizer / Corrector]
                        |
                        v
                  [Generator] --> (retry with corrected prompt)
```

□□□□ 23 (□□: `03-self-correction-patterns.md`)

```python
# Pseudocode for evaluator-optimizer loop
class VideoEvaluator:
    def evaluate(self, video, prompt, criteria):
        return {
            "pass": False,
            "scores": {"visual": 0.6, "alignment": 0.4, "audio": 0.8},
            "feedback": [
                {
                    "dimension": "alignment",
                    "issue": "Video shows a lake but prompt specifies ocean",
                    "severity": "major",
                    "suggestion": "Add 'ocean waves, saltwater, beach' to prompt"
                }
            ]
        }

class PromptOptimizer:
    def optimize(self, original_prompt, feedback):
        # Use LLM to rewrite prompt incorporating feedback
        return corrected_prompt
```

□□□□ 24 (□□: `03-self-correction-patterns.md`)

```
Generated Video
     |
     v
[Tier 1: Cheap Checks]  <-- File valid? Resolution correct? Duration match?
     |                       Audio present? Codec correct?
     |
```

```
   PASS? --no--> REJECT (instant, zero cost)
     |
    yes
     |
     v
[Tier 2: Compute Checks] <-- VMAF score, bitrate analysis
     |                        scene complexity metrics
     |
   PASS? --no--> RE-ENCODE or REJECT
     |
    yes
     |
     v
[Tier 3: Perceptual]     <-- Human QA sample, A/B testing
     |
   PASS? --> SHIP or REJECT
```

□□□□ 25 (□□: `03-self-correction-patterns.md`)

```python
from langgraph.graph import StateGraph, END

class VideoState(TypedDict):
    script: str
    prompt: str
    video_path: str
    quality_scores: dict
    feedback: str
    revision_count: int
    max_revisions: int

def generate_video(state: VideoState) -> VideoState:
    # Generate video from prompt
    ...

def evaluate_quality(state: VideoState) -> VideoState:
    # Run quality checks, populate quality_scores
    ...

def should_retry(state: VideoState) -> str:
    if state["revision_count"] >= state["max_revisions"]:
        return "output_best"
    if state["quality_scores"]["overall"] >= 0.7:
        return "pass"
    if state["quality_scores"]["overall"] < 0.3:
        return "fail_hard"
    return "retry"

def correct_prompt(state: VideoState) -> VideoState:
    # Use VLM feedback to rewrite prompt
    state["revision_count"] += 1
    ...
```

```
graph = StateGraph(VideoState)
graph.add_node("generate", generate_video)
graph.add_node("evaluate", evaluate_quality)
graph.add_node("correct", correct_prompt)
graph.add_node("output", output_video)

graph.set_entry_point("generate")
graph.add_edge("generate", "evaluate")
graph.add_conditional_edges("evaluate", should_retry, {
    "pass": "output",
    "retry": "correct",
    "output_best": "output",
    "fail_hard": END
})
graph.add_edge("correct", "generate")
```

□□□□ 26 (□□: `03-self-correction-patterns.md`)

```
Here are calibration examples:
- [Video A description]: This is a GOOD example because...
- [Video B description]: This is a POOR example because...
- [Video C description]: This is ACCEPTABLE because...

Now evaluate the following video:
```

□□□□ 27 (□□: `03-self-correction-patterns.md`)

```
First describe what you observe in the video.
Then identify any quality issues.
Finally, provide your assessment: GOOD, ACCEPTABLE, or POOR.
```

□□□□ 28 (□□: `03-self-correction-patterns.md`)

```
{
  "observations": "The video shows a dog running...",
  "issues": ["slight motion blur in frames 45-60", "background inconsistency at
0:03"],
  "dimension_scores": {
    "visual_fidelity": "GOOD",
    "content_alignment": "ACCEPTABLE",
    "narrative_coherence": "GOOD"
  },
  "overall": "ACCEPTABLE",
  "corrective_feedback": "Regenerate with emphasis on background stability..."
}
```

□□□□ 29 (□□: `03-self-correction-patterns.md`)

```
Original prompt: "A cat sitting on a windowsill watching rain"

VF-EVAL output:
  - Coherence: GOOD
  - Error: "Cat appears to float slightly above the windowsill"
  - Reasoning: "Physics inconsistency in object placement"

RePrompt output: "A cat sitting firmly on a windowsill, paws resting on
the surface, looking out through a rain-streaked window. Realistic
physics, cat's weight visible on the cushion."
```

□□□□ 30 (□□: `05-gap-analysis.md`)

```
                    Self-Evaluation Capability
                    ^
                    |
          HIGH      |   [Academic Research]
                    |     VISTA, VF-EVAL,
                    |     VideoAgent, FilmAgent
                    |        *
                    |
                    |              +========================+
                    |              |                        |
          MEDIUM    |              |   OPPORTUNITY WINDOW   |
                    |              |                        |
                    |              |   Production system    |
                    |              |   with evaluation +    |
                    |              |   AI video generation  |
                    |              |                        |
                    |              +========================+
                    |
                    |                         [Commercial]
          LOW       |   [Open Source Tools]    Runway, Kling,
                    |    MoneyPrinter,         Synthesia
                    |    ShortGPT, etc.          *
                    |       *
                    |
                    +-------+--------+--------+--------+----> AI Video Integration
                          NONE     BASIC    PARTIAL   FULL

    Legend:
      * = Cluster of existing solutions
      [Box] = Opportunity window for new system
```

□□□□ 31 (□□: `05-gap-analysis.md`)

```
Generation Capability     [============================] High
                          (Open-Sora, Kling API, Runway API)

Assembly Capability       [========================]     High
                          (FFmpeg, MoviePy, Remotion)

Script Generation         [==========================]   High
                          (GPT-4, Claude, DeepSeek)

TTS Quality               [=========================]    High
                          (ElevenLabs, Azure, Edge-TTS)

Evaluation Capability     [====]                         Very Low
                          (No pipeline has any)

Correction Capability     []                             Zero
                          (None implemented anywhere)
```

□□□□ 32 (□□: `06-architecture-proposal.md`)

```
+-------------------------------------------------------------------+
|                         USER INPUT                                |
|  (Topic / Script / Reference images / Style preferences)          |
+------------------------------+------------------------------------+
                               |
                               v
+------------------------------+------------------------------------+
|                      ORCHESTRATOR (LangGraph)                     |
|                                                                   |
|  +-----------+    +-----------+    +------------+    +----------+ |
|  | Script    |--->| Prompt    |--->| Generation |--->| Assembly | |
|  | Generator |    | Engineer  |    | Dispatcher |    | Engine   | |
|  +-----------+    +-----------+    +------------+    +----------+ |
|                                          |                  |      |
|                                     [AI Video API]     [FFmpeg/    |
|                                     Veo 3.1 / Kling   MoviePy]    |
|                                     / Runway / Sora                |
+------------------------------+------------------------------------+
                               |
                               v
+------------------------------+------------------------------------+
|                    QUALITY GATE (3 Layers)                        |
|                                                                   |
|  +------------------+  +------------------+  +-----------------+  |
|  | Layer 1          |  | Layer 2          |  | Layer 3         |  |
|  | Programmatic     |  | ML Model         |  | VLM Review      |  |
|  | Checks           |  | Checks           |  |                 |  |
|  | (<1s, zero cost) |  | (1-10s, low cost)|  | (5-30s, $$)     |  |
|  +--------+---------+  +--------+---------+  +--------+--------+  |
|           |                      |                     |          |
```

```
|        PASS/FAIL             PASS/FAIL/             PASS/FAIL/   |
|                              UNCERTAIN              FEEDBACK     |
+------------------------------+-----------------------------------+
                               |
                   +----------+-----------+
                   |                      |
                 PASS                   FAIL
                   |                      |
                   v                      v
            +------+------+    +-----------+-----------+
            | Final Output |    | CORRECTION AGENT      |
            | (MP4 + JSON  |    |                       |
            |  sidecar)    |    | Feedback --> Corrected|
            +--------------+    | prompt / parameters   |
                                +-----------+-----------+
                                            |
                                            v
                                [Retry: back to Generation]
                                (up to MAX_REVISIONS)
```

□□□□ 33 (□□: `06-architecture-proposal.md`)

```python
import subprocess
import json
from dataclasses import dataclass

@dataclass
class L1Result:
    passed: bool
    checks: dict  # check_name -> {passed: bool, value: any, threshold: any}
    failures: list[str]

def layer1_check(video_path: str, config: dict) -> L1Result:
    """Run all Layer 1 programmatic checks."""
    checks = {}
    failures = []

    # File integrity
    try:
        probe = subprocess.run(
            ["ffprobe", "-v", "quiet", "-print_format", "json",
             "-show_format", "-show_streams", video_path],
            capture_output=True, text=True, timeout=10
        )
        info = json.loads(probe.stdout)
        checks["file_integrity"] = {"passed": True}
    except Exception as e:
        checks["file_integrity"] = {"passed": False, "error": str(e)}
        failures.append("file_integrity")
        return L1Result(passed=False, checks=checks, failures=failures)
```

```python
# Resolution check
video_stream = next(
    (s for s in info["streams"] if s["codec_type"] == "video"), None
)
if video_stream:
    w, h = int(video_stream["width"]), int(video_stream["height"])
    target_w, target_h = config["target_resolution"]
    res_ok = (w == target_w and h == target_h)
    checks["resolution"] = {
        "passed": res_ok, "value": f"{w}x{h}",
        "threshold": f"{target_w}x{target_h}"
    }
    if not res_ok:
        failures.append("resolution")

# Duration check
duration = float(info["format"]["duration"])
target_dur = config["target_duration"]
tolerance = config.get("duration_tolerance", 0.1)
dur_ok = abs(duration - target_dur) / target_dur <= tolerance
checks["duration"] = {
    "passed": dur_ok, "value": duration,
    "threshold": f"{target_dur} +/- {tolerance*100}%"
}
if not dur_ok:
    failures.append("duration")

# Audio stream check
audio_stream = next(
    (s for s in info["streams"] if s["codec_type"] == "audio"), None
)
checks["audio_present"] = {"passed": audio_stream is not None}
if not audio_stream:
    failures.append("audio_present")

# A/V sync check (simplified: compare start times)
if video_stream and audio_stream:
    v_start = float(video_stream.get("start_time", 0))
    a_start = float(audio_stream.get("start_time", 0))
    sync_diff_ms = abs(v_start - a_start) * 1000
    sync_ok = sync_diff_ms < config.get("max_av_sync_ms", 100)
    checks["av_sync"] = {
        "passed": sync_ok, "value": f"{sync_diff_ms:.1f}ms",
        "threshold": f"<{config.get('max_av_sync_ms', 100)}ms"
    }
    if not sync_ok:
        failures.append("av_sync")

# Audio loudness check
loudness_result = subprocess.run(
    ["ffmpeg", "-i", video_path, "-af", "loudnorm=print_format=json",
     "-f", "null", "-"],
```

```
        capture_output=True, text=True, timeout=30
    )
    # Parse loudness from stderr (FFmpeg outputs to stderr)
    # ... (parsing logic for integrated loudness)

    return L1Result(
        passed=len(failures) == 0,
        checks=checks,
        failures=failures
    )
```

□□□□ 34 (□□: `06-architecture-proposal.md`)

```
def compute_l2_score(checks: dict, weights: dict) -> float:
    """
    Compute weighted Layer 2 quality score.

    Default weights (configurable via quality_thresholds.yaml):
      dover_overall: 0.25
      clip_alignment: 0.25
      subject_consistency: 0.15
      temporal_flickering: 0.15
      motion_smoothness: 0.10
      color_consistency: 0.05
      subtitle_sync: 0.05
    """
    score = 0.0
    for check_name, weight in weights.items():
        if check_name in checks:
            score += checks[check_name] * weight
    return score
```

□□□□ 35 (□□: `06-architecture-proposal.md`)

```
@dataclass
class L2Result:
    score: float
    dimension_scores: dict  # dimension -> float
    decision: str  # "pass", "uncertain", "fail"
    details: dict

def layer2_check(
    video_path: str, prompt: str, config: dict
) -> L2Result:
    """Run Layer 2 ML model checks."""
    scores = {}

    # DOVER quality assessment
    dover_scores = dover_model.evaluate(video_path)
```

```python
    scores["dover_overall"] = dover_scores["overall"]
    scores["dover_aesthetic"] = dover_scores["aesthetic"]
    scores["dover_technical"] = dover_scores["technical"]

    # OpenCLIP text-video alignment
    frames = extract_frames(video_path, n_frames=16)
    clip_score = compute_clip_similarity(frames, prompt)
    scores["clip_alignment"] = clip_score

    # VBench subset
    vbench_results = vbench_evaluator.evaluate(
        video_path, prompt,
        dimensions=["subject_consistency", "temporal_flickering",
                    "motion_smoothness"]
    )
    scores.update(vbench_results)

    # Scene detection
    scenes = detect_scenes(video_path)
    expected_scenes = config.get("expected_scenes", len(scenes))
    scores["scene_consistency"] = 1.0 if len(scenes) <= expected_scenes + 1 else 0.5

    # Aggregate
    weights = config.get("l2_weights", DEFAULT_L2_WEIGHTS)
    overall = compute_l2_score(scores, weights)

    # Decision
    if overall >= config.get("l2_pass_threshold", 0.70):
        decision = "pass"
    elif overall >= config.get("l2_uncertain_threshold", 0.50):
        decision = "uncertain"
    else:
        decision = "fail"

    return L2Result(
        score=overall,
        dimension_scores=scores,
        decision=decision,
        details={"weights": weights}
    )
```

□□□□ 36 (□□: `06-architecture-proposal.md`)

```python
def sample_frames_for_vlm(video_path: str, n_frames: int = 16) -> list:
    """
    Uniform temporal sampling of frames for VLM input.

    16 frames balances coverage vs token cost:
    - Too few (<8): miss temporal issues
    - Too many (>24): diminishing returns, higher token cost
```

```
    """
    cap = cv2.VideoCapture(video_path)
    total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    indices = np.linspace(0, total_frames - 1, n_frames, dtype=int)
    frames = []
    for idx in indices:
        cap.set(cv2.CAP_PROP_POS_FRAMES, idx)
        ret, frame = cap.read()
        if ret:
            frames.append(frame)
    cap.release()
    return frames
```

□□□□ 37 (□□: `06-architecture-proposal.md`)

```
Labels: EXCELLENT / GOOD / FAIR / POOR
```

□□□□ 38 (□□: `06-architecture-proposal.md`)


```
{
  "observations": "The video shows a sequence of 5 shots depicting...",
  "dimension_assessments": {
    "visual_fidelity": {
      "reasoning": "Frame quality is consistent. Minor aliasing on text overlay in
shot 3. No major artifacts.",
      "label": "GOOD",
      "issues": ["minor text aliasing in shot 3"]
    },
    "content_alignment": {
      "reasoning": "Script mentions 'a busy coffee shop' but shots 2-3 show an empty
room.",
      "label": "FAIR",
      "issues": ["shots 2-3 do not match 'busy coffee shop' description"]
    },
    "narrative_coherence": {
      "reasoning": "Shots flow logically. Transition from intro to body to
conclusion is clear.",
      "label": "GOOD",
      "issues": []
    },
    "overall_feel": {
      "reasoning": "Professional-looking but the content mismatch in shots 2-3 is
distracting.",
      "label": "FAIR",
      "issues": ["content mismatch reduces overall quality"]
    }
  },
  "overall_label": "FAIR",
  "corrective_feedback": {
```

```
    "primary_issue": "Shots 2-3 show an empty room instead of a busy coffee shop",
    "suggested_prompt_changes": "Replace shots 2-3 with: 'Interior of a busy coffee
shop, customers at tables, barista making drinks, warm lighting, afternoon'",
    "parameter_adjustments": {
      "shots_to_regenerate": [2, 3],
      "style_guidance": "Increase detail specificity for indoor scenes"
    }
  }
}
```

□□□□ 39 (□□: `06-architecture-proposal.md`)

```python
def reprompt(original_prompt: str, feedback: dict) -> str:
    """Use LLM to generate corrected prompt based on VLM feedback."""
    system = """You are a video generation prompt engineer. Given the
    original prompt and quality feedback, rewrite the prompt to address
    the identified issues. Be specific and concrete."""

    user = f"""Original prompt: {original_prompt}

    Issue: {feedback['primary_issue']}
    Suggested changes: {feedback['suggested_prompt_changes']}

    Rewrite the prompt to fix this issue while preserving the original intent."""

    corrected = llm.generate(system=system, user=user)
    return corrected
```

□□□□ 40 (□□: `06-architecture-proposal.md`)

```python
from typing import TypedDict, Literal
from langgraph.graph import StateGraph, END

class VideoState(TypedDict):
    # Input
    topic: str
    script: str
    target_duration: float
    style_config: dict

    # Generation
    shot_prompts: list[str]
    video_clips: list[str]  # file paths
    assembled_video: str     # file path

    # Evaluation
    l1_result: dict | None
    l2_result: dict | None
    l3_result: dict | None
```

```python
    quality_scores: dict

    # Correction
    feedback: str
    corrective_actions: list[dict]
    revision_count: int
    max_revisions: int
    retry_strategy: str  # "cheap", "correction", "degradation"

    # Output
    final_video: str
    quality_report: dict
    warnings: list[str]


def generate_script(state: VideoState) -> VideoState:
    """Generate or refine the video script from topic."""
    # ... LLM script generation
    return state


def generate_shots(state: VideoState) -> VideoState:
    """Generate individual video clips from shot prompts."""
    # ... AI video API calls (Runway/Kling/Sora)
    return state


def assemble_video(state: VideoState) -> VideoState:
    """Assemble clips + audio + subtitles into final video."""
    # ... FFmpeg/MoviePy assembly
    return state


def quality_gate(state: VideoState) -> VideoState:
    """Run three-layer quality gate."""
    # Layer 1
    l1 = layer1_check(state["assembled_video"], state["style_config"])
    state["l1_result"] = l1.__dict__
    if not l1.passed:
        state["retry_strategy"] = "cheap"
        return state

    # Layer 2
    l2 = layer2_check(
        state["assembled_video"],
        state["script"],
        state["style_config"]
    )
    state["l2_result"] = l2.__dict__
    if l2.decision == "pass":
        state["quality_scores"] = l2.dimension_scores
        return state
```

```python
    elif l2.decision == "fail":
        state["retry_strategy"] = "cheap"
        return state

    # Layer 3 (only if L2 uncertain)
    l3 = vlm_review(state["assembled_video"], state["script"])
    state["l3_result"] = l3
    state["quality_scores"] = {**l2.dimension_scores, "vlm": l3}

    if l3["overall_label"] in ("EXCELLENT", "GOOD"):
        state["retry_strategy"] = "none"
    elif l3["overall_label"] == "FAIR":
        state["retry_strategy"] = "correction"
        state["feedback"] = json.dumps(l3["corrective_feedback"])
    else:  # POOR
        state["retry_strategy"] = "correction"
        state["feedback"] = json.dumps(l3["corrective_feedback"])

    return state


def should_retry(state: VideoState) -> str:
    """Determine next step based on quality gate result."""
    strategy = state.get("retry_strategy", "none")

    if strategy == "none":
        return "output"

    if state["revision_count"] >= state["max_revisions"]:
        return "degrade"

    if strategy == "cheap":
        return "retry_cheap"

    if strategy == "correction":
        return "retry_correct"

    return "output"


def cheap_retry(state: VideoState) -> VideoState:
    """Retry with same prompts but new random seed."""
    state["revision_count"] += 1
    # Regenerate with new seed
    return state


def correction_retry(state: VideoState) -> VideoState:
    """Use VLM feedback to correct prompts, then regenerate."""
    state["revision_count"] += 1
    feedback = json.loads(state["feedback"])
```

```python
    # Identify which shots to regenerate
    shots_to_redo = feedback.get("parameter_adjustments", {}).get(
        "shots_to_regenerate", list(range(len(state["shot_prompts"])))
    )

    # Rewrite prompts for failed shots
    for shot_idx in shots_to_redo:
        state["shot_prompts"][shot_idx] = reprompt(
            state["shot_prompts"][shot_idx], feedback
        )

    return state


def degrade_output(state: VideoState) -> VideoState:
    """Output best version with quality warning."""
    state["warnings"].append(
        f"Quality target not met after {state['revision_count']} revisions. "
        f"Outputting best available version."
    )
    return state


def output_video(state: VideoState) -> VideoState:
    """Finalize output with quality report sidecar."""
    state["final_video"] = state["assembled_video"]
    state["quality_report"] = {
        "l1": state.get("l1_result"),
        "l2": state.get("l2_result"),
        "l3": state.get("l3_result"),
        "revisions": state["revision_count"],
        "warnings": state["warnings"]
    }
    return state


# Build the graph
graph = StateGraph(VideoState)

graph.add_node("generate_script", generate_script)
graph.add_node("generate_shots", generate_shots)
graph.add_node("assemble", assemble_video)
graph.add_node("quality_gate", quality_gate)
graph.add_node("cheap_retry", cheap_retry)
graph.add_node("correction_retry", correction_retry)
graph.add_node("degrade", degrade_output)
graph.add_node("output", output_video)

graph.set_entry_point("generate_script")
graph.add_edge("generate_script", "generate_shots")
graph.add_edge("generate_shots", "assemble")
graph.add_edge("assemble", "quality_gate")
```

```
graph.add_conditional_edges("quality_gate", should_retry, {
    "output": "output",
    "retry_cheap": "cheap_retry",
    "retry_correct": "correction_retry",
    "degrade": "degrade",
})

graph.add_edge("cheap_retry", "generate_shots")
graph.add_edge("correction_retry", "generate_shots")
graph.add_edge("degrade", "output")
graph.add_edge("output", END)

pipeline = graph.compile()
```

□□□□ 41 (□□: `06-architecture-proposal.md`)

```
RETRY_BUDGETS = {
    "stock_footage": {
        "max_revisions": 5,
        "l1_max_retries": 3,    # cheap, just re-select footage
        "l2_max_retries": 2,    # ML re-evaluation
        "l3_max_reviews": 1,    # VLM review is expensive relative to stock
    },
    "ai_video_api": {
        "max_revisions": 3,
        "l1_max_retries": 2,
        "l2_max_retries": 1,
        "l3_max_reviews": 1,
    },
    "local_model": {
        "max_revisions": 2,
        "l1_max_retries": 1,
        "l2_max_retries": 1,
        "l3_max_reviews": 1,
    }
}
```

□□□□ 42 (□□: `06-architecture-proposal.md`)

```
# Quality Gate Configuration
# --------------------------

layer1:
  checks:
    file_integrity: true
    resolution: true
    duration: true
    av_sync: true
```

```yaml
    audio_loudness: true
    subtitle_validation: true
    frame_consistency: true
    audio_present: true
    codec_validation: true

  thresholds:
    max_av_sync_ms: 100
    target_loudness_lufs: -23.0
    loudness_tolerance_db: 2.0
    duration_tolerance_pct: 0.10
    frame_hash_threshold: 15

layer2:
  weights:
    dover_overall: 0.25
    clip_alignment: 0.25
    subject_consistency: 0.15
    temporal_flickering: 0.15
    motion_smoothness: 0.10
    color_consistency: 0.05
    subtitle_sync: 0.05

  thresholds:
    pass: 0.70         # >= this: PASS
    uncertain: 0.50    # >= this but < pass: escalate to L3
    # < uncertain: FAIL + cheap retry

layer3:
  vlm:
    provider: gemini       # gemini | openai | anthropic
    model: gemini-2.5-flash
    fallback_provider: openai
    fallback_model: gpt-4o

  frame_sampling:
    n_frames: 16
    method: uniform        # uniform | keyframe | adaptive

  scoring:
    labels: [EXCELLENT, GOOD, FAIR, POOR]
    label_values:
      EXCELLENT: 1.0
      GOOD: 0.8
      FAIR: 0.5
      POOR: 0.2
    dimensions:
      visual_fidelity: 0.30
      content_alignment: 0.30
      narrative_coherence: 0.25
      overall_feel: 0.15
```

```yaml
  anchor_samples:
    good: "examples/good_sample.mp4"
    poor: "examples/poor_sample.mp4"

retry:
  max_revisions: 3
  strategies:
    cheap:
      trigger: "l1_fail OR l2_score < 0.50"
      action: "regenerate with new seed"
    correction:
      trigger: "l2_score 0.50-0.69 OR l3_label FAIR/POOR"
      action: "vlm feedback -> reprompt -> regenerate"
    degradation:
      trigger: "revision_count >= max_revisions"
      action: "output best + warning"
    escalation:
      trigger: "3x consecutive POOR"
      action: "stop + notify human"
```

□□□□ 43 (□□: `06-architecture-proposal.md`)

```yaml
# Pipeline Configuration
# ----------------------

generation:
  providers:
    primary:
      name: kling
      api_key_env: KLING_API_KEY
      model: kling-v3
      default_resolution: "1080x1920"
      default_duration: 5
      timeout_seconds: 120
    fallback:
      name: runway
      api_key_env: RUNWAY_API_KEY
      model: gen-4.5
      default_resolution: "1080x1920"
      default_duration: 5
      timeout_seconds: 180

  concurrency:
    max_parallel_generations: 4
    rate_limit_per_minute: 10

script:
  llm:
    provider: openai     # openai | anthropic | deepseek
    model: gpt-4o
```

```yaml
  language: en
  max_shots: 12
  target_duration_seconds: 60

tts:
  provider: edge-tts     # edge-tts | elevenlabs | azure | openai
  voice: en-US-ChristopherNeural
  speed: 1.0
  fallback_provider: openai

subtitles:
  method: stable-ts      # stable-ts | whisperx | whisper
  model_size: large-v3
  word_level: true
  style:
    font: "Montserrat Bold"
    font_size: 48
    color: "#FFFFFF"
    outline_color: "#000000"
    outline_width: 2
    position: bottom     # bottom | center

bgm:
  source: royalty-free   # royalty-free | suno | musicgen
  library_path: "assets/bgm/"
  volume_db: -18         # relative to narration

output:
  format: mp4
  video_codec: h264
  audio_codec: aac
  video_bitrate: "8M"
  audio_bitrate: "192k"
  fps: 30
  sidecar: true          # output quality report JSON alongside video
```

□□□□ 44 (□□: `06-architecture-proposal.md`)

```
LangGraph Pipeline
    │
    ├── generate_shots ──trace──▶ LangSmith
    │       └── [provider, prompt, latency, cost]
    │
    ├── quality_gate ──trace──▶ LangSmith
    │       ├── L1 results [check_name → pass/fail]
    │       ├── L2 results [dimension_scores, aggregate]
    │       └── L3 results [VLM response, label, critique]
    │
    ├── correction_retry ──trace──▶ LangSmith
    │       └── [original_prompt, corrected_prompt, diff]
    │
```

```
    └── output ──trace──▶ LangSmith
            └── [final_scores, retry_count, warnings]
```

□□□□ 45 (□□: `06-architecture-proposal.md`)

```python
# Add to orchestrator.py
import os
os.environ["LANGCHAIN_TRACING_V2"] = "true"
os.environ["LANGCHAIN_API_KEY"] = "..."
os.environ["LANGCHAIN_PROJECT"] = "videoqa-gate"

# LangGraph pipeline automatically sends traces to LangSmith
# when LANGCHAIN_TRACING_V2 is set
pipeline = graph.compile()
result = pipeline.invoke(initial_state)
```

□□□□ 46 (□□: `06-architecture-proposal.md`)

```
tests/
  fixtures/
    good_sample.mp4         -- professionally produced, all checks should pass
    bad_resolution.mp4      -- wrong resolution
    bad_sync.mp4            -- A/V sync drift
    low_quality.mp4         -- heavy compression artifacts
    no_audio.mp4            -- video only
    flickering.mp4          -- temporal flickering artifacts
    mismatched_content.mp4  -- video does not match prompt
```

□□□□ 47 (□□: `06-architecture-proposal.md`)

```python
# google-genai SDK async polling pattern
from google import genai
from google.genai import types

client = genai.Client(api_key=GEMINI_API_KEY)

operation = client.models.generate_videos(
    model="veo-3.1-generate-preview",
    prompt="Cinematic shot of...",
    config=types.GenerateVideosConfig(
        aspect_ratio="9:16",    # Vertical short video
        number_of_videos=1,
    ),
)

# Poll → Download → Save
while not operation.done:
    time.sleep(5)
```

```
    operation = client.operations.get(operation)

video = operation.response.generated_videos[0]
client.files.download(file=video.video)
video.video.save("output.mp4")
```

□□□□ 48 (□□: `07-workflow-analysis.md`)

```
□□ 6 □□2025□□    □□ → TTS → □□□□ → □□ → □□ → □□
□□ 4 □□2026.02+□□□□ → AI □□+□□□□□□ → □□ → □□
```

□□□□ 49 (□□: `07-workflow-analysis.md`)

```
LLM □□ → edge-tts → Pexels □□(CLIP □□□□) → Whisper □□ → BGM → FFmpeg
                                ↑
                □□□□□ → CLIP □□□□□□□ #1 □□□□□
```

□□□□ 50 (□□: `07-workflow-analysis.md`)

```
LLM □□(□□□□□) → Kling 3.0(□□□□□+□□) → Whisper □□ → FFmpeg □□
```

□□□□ 51 (□□: `07-workflow-analysis.md`)

```
LLM □□□□□ → Kling 3.0 Pro(4K/60fps □□□+□□) → Whisper □□ → FFmpeg
```

□□□□ 52 (□□: `07-workflow-analysis.md`)

```
2026.01.05  LTX-2 □□□□□4K/50fps+□□□Apache 2.0□
2026.01.07  Sora □□□□□□
2026.01.10  Sora □□□□□□□
2026.02.05  ★ Kling 3.0 □□ — □□□□ 4K/60fps AI □□□□
2026.02.07  ★ Seedance 2.0 □□ — □□□□ + 12 □□□□□□□
2026.02.10  ★ Runway Gen-4.5 API □□
2026.02.12  Seedance 2.0 □□□□□□
2026.03.04  Seedance 2.0 API □□□□□□□□□□□□□□
2026.03.xx  NVIDIA GDC □□ ComfyUI App View + LTX-2.3 □□
```

□□□□ 53 (□□: `07-workflow-analysis.md`)

```
2025 □□□                        2026.02 □□□□□
————————————                      ————————————

□□□□□                           → □□□□□□□□
□□/□□□□□□                      → □□□□□□□□□□
1080p □□□                          → 4K/60fps □□□□
□□/□□□□                         → □□□□□□12+ □□□□□
```

```
□□□ prompt □□                    → □□□ + □□□□□□□□
□□□□□□□                          → □□□□□□□□□□□□+□□+□□□
```

□□□□ 54 (□□: `07-workflow-analysis.md`)

```
Runway Gen-4     ▓▓▓
```

```
□□□□□□□MoneyPrinterTurbo □□: #1 □□□□ > #2 □□□□□ > #7 □□□□
AI □□□□Veo/Runway/Kling□:     #3 □□□□□ > #4 □□□□ > #6 □□□□□
□□/□□□□HeyGen/Synthesia□:   #5 □□□□ > #8 □□□□ > #9 AI □□
```

□□□□ 62 (□□: `07-workflow-analysis.md`)

```
□□□□□□□□□□□:                    □□□□□□□□□□□:
────────────────                ────────────────
□ □□□□□□LLM□                     □ □□□□□□□□□□□□□□□□
□ TTS□Edge-TTS □□□□□             □ □□□□□□□□□□/□□
□ □□□Whisper□                    □ □□□□□ Prompt □□
□ □□□FFmpeg□                     □ □□-□□□□□□□□□□→CLIP□
□ AI □□□□□□□□□□□□            □ □□□□□/□□□□□
```

□□□□ 63 (□□: `07-workflow-analysis.md`)

```
□□ 1□□□□□□□□□□□□
  → □ 100 □□□□□□□□□ L1+L2 □□□□□□□□□□□□
  → □□□□□□□□□□□□

□□ 2□□□ B□□□+□□□
  → □□□ N=2 □□ → □□□□□
  → □□□□□□ "prompt→□□→□□" □□□□□

□□ 3□□□ A□□□□□□□□
  → □□□ 2 □□□□ prompt □□
  → □□□□□□□□□□□□□□□□
```

□□□□ 64 (□□: `08-best-practices.md`)

```
DeepSeek □□□ → Edge-TTS □□ → Pexels □□ → CapCut □□□□ → YouTube Audio Library □□ →
CapCut □□
```

□□□□ 65 (□□: `08-best-practices.md`)

```
GPT-4o-mini □□□ → Fish Audio □□ → Kling 3.0 □□□□□□□ → CapCut □□□□+□□ → BGM → □□
```

□□□□ 66 (□□: `08-best-practices.md`)

```
□□□□ → □□□□□□□□+□□+Pexels□□+□□+BGM+□□□→ □□□□ → □□
```

□□□□ 67 (□□: `08-best-practices.md`)

```
□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□ {30/60} □□□□□□□□
```

```
□□□{topic}
□□□{□□/TikTok/YouTube Shorts}
□□□{□□□□/□□/□□/□□}

□□□
1. □□□□ Hook(0-3s)□Build□Body□CTA □□□
2. Hook □□□□ 3 □□□□□□□□□□{□□□/□□/□□/□□□□}□□
3. □□□□□□□[□□□□]□□□□□□
4. □□□□□□□□□□□□□□□
5. □□□□□ {150-200} □

□□□□□
□□1 [0-3s] Hook□...
□□2 [3-10s] Build□...
□□3 [10-25s] Body□...
□□4 [25-30s] CTA□...
```

□□□□ 68 (□□: `08-best-practices.md`)

```
□ 1 □□□□□□□□AI□2 □□□
  ├── □□□ChatGPT / □□ / DeepSeek
  ├── □□□□□ + Prompt □□□□□ 1 □□
  └── □□□□□□□□□□□□□□□□

□ 2 □□□□□TTS□1 □□□
  ├── □□□Edge-TTS□□□□□ Fish Audio□$5.5/□□
  ├── □□□□□ 1.0-1.15x□□□□□□□□□
  └── □□□narration.mp3

□ 3 □□□□□□□□AI/□□□□5 □□□
  ├── □□ A□Pexels □□ + CLIP □□□□□□$0□
  ├── □□ B□Kling 3.0 □□□□□□□$0□□□ 66 □□□
  ├── □□ C□MidJourney/DALL-E □□□□□~$0.05/□□
  └── □□□□□□□□□□□□□/□□□□

□ 4 □□□□□□□□CapCut/□□□5 □□□
  ├── □□□□ + □□
  ├── □□□□□
  ├── □□□□□□□ OK □□
  ├── □□ BGM□□□□ ducking □ 15-20%□
  ├── □□□□□□
  └── □□□□□□□□

□ 5 □□□□□□□□2-3 □□□
  ├── □□□ 3 □□□□□□□□
  ├── □□□□□□□□□
  ├── □□□□/□□
  ├── □□□□
  └── □□□□□□□□

□ 6 □□□□□□1 □□□
```

```
  ├── □□□□□□□□
  ├── □ SEO □□□□□/□□□□
  ├── □□□□□□□□□
  └── □□
```

□□□□ 69 (□□: `08-best-practices.md`)

```
□ 1 □□□□□□□
  └── Web UI □□□□/□□□

□ 2 □□□□□□□
  ├── LLM □□□□OpenAI/DeepSeek/Gemini/Ollama□
  ├── TTS □□□Edge-TTS/GPT-SoVITS/Azure□
  ├── □□□□□9:16 □ 16:9□
  ├── □□□□□□□/□□/□□/□□□
  └── BGM □□□□□/□□/□□□

□ 3 □□□□□□□
  └── □□□□□□□□□□→□□→Pexels □□→□□→BGM→FFmpeg □□

□ 4 □□□□□□□□2-3 □□□
  ├── □□□□□□□□□□□□□
  ├── □□□□□□□□□□□
  └── □□□□□□□□□□□

□ 5 □□□□
```

□□□□ 70 (□□: `08-best-practices.md`)

```
n8n/Make □□□□□□□
  ├── Trigger□Notion/Airtable □□□□
  ├── Step 1□LLM□Claude/GPT-4o□→ □□ + Prompt □□
  ├── Step 2□AI □□□Kling/Runway/Seedance via API□→ □□□□
  ├── Step 3□TTS□ElevenLabs/Fish Audio□→ □□
  ├── Step 4□FFmpeg□custom code node□→ □□ + □□□□□□
  ├── Step 5□Whisper.cpp → □□□□
  └── Step 6□□□ API → □□□□□□TikTok/Instagram/YouTube/Facebook□
```

□□□□ 71 (□□: `08-best-practices.md`)

```
□ 1 □□□□□□□□ (Agent)
  └── □ Coze □ Dify □□□□□□□□□□“□□□□□□□□□”□

□ 2 □□□□□□□□□□
  ├── □□□□□□□□□□□□□□□□
  ├── □□□□□□□□□□ DeepSeek □□ 5 □□□□
  ├── □□□□□□□□□□□□ API (Midjourney) □ □□□ API (Seedance)□
  ├── □□□□□□□□ TTS API□
  └── □□□□□□□□□□□□□□□ MP4 □□□□□
```

```
□ 3 □□□□□□□□
  └── □□□□□□□□“□□□□□□□□□□□□□□□□□□□□□□□”□Agent □□□□□□□□□
```

□□□□ 72 (□□: `08-best-practices.md`)

```
□ 1 □□□□□□□□
  └── LLM □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□ 2 □□Kling/Seedance □□□□□
  ├── □ prompt □□□□□□□□ / □□□□□□□
  ├── □□□□ 2-6 □□□□□ + □□□□
  ├── □□ Subject Reference (`@`□□□□) □□□□□
  └── □□□□□□□□□□□□□□+□□

□ 3 □□□□
  └── □□□□□□□□□□ OK □□□
```

□□□□ 73 (□□: `09-commercial-production-guide.md`)

```
□□         □□          □□                  □□          □□         □□          □□
□□
 |            |            |                     |            |            |
|            |
Claude    Claude/GPT-4o   Runway Gen-4.5        ElevenLabs   CapCut      Musicbed
Premiere    □□□
 □□        □□□□□    / Seedance 2.0 (□□) □□□       □□OK      □□□□      □□□□     □□□
 □□        + Prompt      Veo 3.1 (□□/□□)    □□□□     □□+□□   /Artlist      +□□       □
□
             □□         (□□□□□□□□)                    □□                     DaVinci
```

□□□□ 74 (□□: `09-commercial-production-guide.md`)

```
□□□□□□□□□□□□□□□□□□□
```

```
□□□□□ Markdown□□□□□□□□□□□□□□□□□□□□□□□Hook□□□□□□□□□
```

□□□□ 75 (□□: `09-commercial-production-guide.md`)

```
□□□□□□□□□□□□□□ {30} □□□□□□□□□□□□□

□□□□□□{□ 1 □□□□□□□}
□□/□□□{□□□□} - {□□}
□□□□□□{□□□□/□□□□□/□□□□□/□□□□□}

□□□□□□
1. □□□□□□□□□□□□□□□□□□□□□□□□□□□□
   [00:00-00:03] Hook — □□□□□ | □□□□□ | □□/□□□□□
   [00:03-00:08] Build — □□
   [00:08-00:23] Body — □□□□□□□□□□□□□
   [00:23-00:30] CTA — □□

2. □□□□□□□□□
   - □□□□□□□□□□□□□□□/□□/□□/□□/□□□□□□
   - □□□□□□□□□□□□□□□□□□□□□□ AI □□□□□□□60-80 words□
   - □□□□□□□□□□□□□
   - □□□□□□□□□/□□□□□
   - □□□□□□□□□□□□□□□□

3. □□□□□□□□□□□□□□□□□□□□□□□□□□□
   - □□□□□□□□□□□□
   - □□□□□□□□□□□ "warm golden hour lighting"□
   - □□□□□□□□ "slow dolly forward"□
   - □□□□□□□□□ "shallow depth of field, focus on product"□

4. □□□□□□□□□□□□□□□□□□□□
```

□□□□ 77 (□□: `09-commercial-production-guide.md`)

```
□ 1. □□□15 □□□
  □ Claude □□□□ Brief□□□ 3 □□□□□
  □ □□□□□□□

□ 2. □□□20 □□□
  □ Claude/GPT-4o □□□□□□□□□□□□□□□□□□
  □ □□□□□□□/□□/□□/□□

□ 3. □□□□□40 □□□
  □ □□□□□MidJourney □□□□□□□
  □ Runway Gen-4.5 Image-to-Video□□□□ 2-3 □□□
  □ □□□□□
  □ □□□□□□□□□□ Veo 3.1

□ 4. □□□10 □□□
  □ ElevenLabs □□□□□□□□□□□
  □ □□□□/□□

□ 5. □□□15 □□□
  □ CapCut/Premiere □□□□□□
  □ □□□□□
  □ □□□□ + □□□□/□□
  □ BGM + □□

□ 6. □□□15 □□□
  □ □□□□□□□□□□□□□□
  □ □□□□
  □ Logo/□□□□
  □ □□□□□ -10dBFS□BGM -24dBFS□
  □ □□□□□□□

□ 7. □□□5 □□□
  □ □□□□□□□□□□□
  □ □□□□□□/□□/□□
```

□□□□ 78 (□□: `09-commercial-production-guide.md`)

```
AI □□□□□□□□□□□          □□□□□□□□□□□□□
├── □□/□□□□□□□           ├── □□□□
├── □□/□□□□                ├── □□/□□□□□
├── □□/□□□□                ├── □□□□□□
├── □□/□□□□□              └── □□ Logo □□
└── □□/□□□□
```

□□□□ 79 (□□: `10-ultimate-guide.md`)

```
① DeepSeek □□□ (Hook-Build-Body-CTA)
     ↓
② Edge-TTS □□□zh-CN-XiaoxiaoNeural□
     ↓
③ Pexels □□□□ + CLIP □□□□□□□□□/Kling □□□□□□
     ↓
④ CapCut □□ → □□□□ OK □□ → BGM□Pixabay□→ □□
     ↓
□ □□□□□3 □□□□□□□□□□□□□□□
     ↓
□ □□ 9:16 → □□
```

□□□□ 80 (□□: `10-ultimate-guide.md`)

```
① LLM □□□□□□□□□□□□□□□□□□□□□□□□□□
     ↓
② MidJourney □□□□□□□□□□□□□□□□
     ↓
③ Runway/Seedance/Veo Image-to-Video□□□□ 5-8 □□2-3 □□□□□□
     ↓
④ Whisper □□ → CapCut □□ OK □□
     ↓
⑤ Premiere/DaVinci □□□□□+□□+Logo□
     ↓
□ □□□□ → □□□□□ → □□
```

□□□□ 81 (□□: `10-ultimate-guide.md`)

```
Coze/Dify □□□□□□
  ├── □□□□□□□□□□□□□□
  ├── LLM □□□DeepSeek □□ 5 □□□ Prompt
  ├── □□ API □□□
  │   ├── SiliconFlow → Wan 2.2 □□□□
  │   ├── □ Runway/Seedance API → □□□□
  │   └── ElevenLabs/Fish Audio → TTS
  ├── □□□□□FFmpeg □□ + □□
  └── □□□□□MP4 □□□□

□□□□□"□□□□□□□□□□□□"
Agent □□□□□□□□ MP4
```

□□□□ 82 (□□: `10-ultimate-guide.md`)

```
ComfyUI □□□□
  Load Image → Upscale → LTX-2 Video Node → Audio Generation → Concat → Export MP4
```

□□□□ 83 (□□: `10-ultimate-guide.md`)

```
□□□□□□□□□□□□□□□□□□□□□□□ {30/60} □□□□□
□□□{topic}
□□□{□□/TikTok/YouTube Shorts}
□□□{□□□□/□□/□□/□□}

□□□
1. Hook(0-3s) + Build + Body + CTA □□□
2. Hook □□{□□□/□□/□□}□□
3. □□□□□ [□□□□]□□□□60-80 words□
4. □□□ {150-200} □

□□□□□
[00:00-00:03] Hook□□□ | [□□□□] | □□□□
[00:03-00:10] Build□□□ | [□□□□] | □□□□
...
```

□□□□ 84 (□□: `10-ultimate-guide.md`)

```
[□□□□], [□□], [□□], [□□], [□□], [□□], [□□□□]. [□□□□].

□□□
Close-up shot, a woman's hand placing a luxury bottle on marble, warm golden
hour lighting, shallow depth of field with bokeh, slow dolly-in. Cinematic
commercial, anamorphic lens flare.
```

□□□□ 85 (□□: `10-ultimate-guide.md`)

```
□□ + □□ + □□ + □□ + □□ + □□ + □□ + □□

□□□
@□□1 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
□□□□□□□□□□□□□□□□□□□
```

```
pip install edge-tts
edge-tts --text "□□□□" --voice zh-CN-XiaoxiaoNeural --rate "+10%" -o output.mp3
```

□□□□ 88 (□□: `10-ultimate-guide.md`)

```
□□□□+□□ → □□ > □□□□ → □□□ OK □□ → □□□□□/□□ → □□□□ → □□
```

□□□□ 89 (□□: `10-ultimate-guide.md`)

```
# □□□□□ vs □□□□□□□□□□□
ffmpeg -i encoded.mp4 -i original.mp4 \
  -filter_complex libvmaf=model_path=vmaf_v0.6.1.json:log_path=log.json:log_fmt=json
\
  -f null -
```

□□□□ 90 (□□: `10-ultimate-guide.md`)

```
import torch
from torchmetrics.functional.audio import dnsmos
scores = dnsmos(preds=audio_tensor, fs=16000)
# □□ [P808_MOS, MOS_SIG, MOS_BAK, MOS_OVR]
# □□□MOS_OVR >= 3.5□□□□□□□>= 4.0□□□□□
```

□□□□ 91 (□□: `10-ultimate-guide.md`)

```
□ 1 □□□□□□□□□□ GPU□
  └── ComfyUI — □□□□□□□□□□ Wan 2.2 / LTX-2

□ 2 □□□□□□□□□□□□□
  └── Coze / Dify — LLM □□□□□□□□□□

□ 3 □□□□□□□□□□□+□□□
  └── n8n / Make — □□ Notion → □□□□ → □□□□
```

□□□□ 92 (□□: `10-ultimate-guide.md`)

```
1. □□ Bot → □□ Workflow □□
2. Start □□□□□□□□□□
3. LLM □□□DeepSeek □□□□ → □□□□ JSON
4. Plugin □□□□□ Runway/Seedance/SiliconFlow API → □□
5. Plugin □□□□□ TTS API → □□
6. Code □□□FFmpeg □□
7. End □□□□□ MP4 □□□□
```

□□□□ 93 (□□: `10-ultimate-guide.md`)

```
1. □□ SiliconCloud → □□ API Key
2. Dify □□□□□□ SiliconFlow □□
3. □□ Workflow → □□ SiliconFlow Video Generate □□
4. □□□□□Wan-AI/Wan2.2-I2V-A14B
5. □□ prompt + image □□ + video_size (1280x720)
6. End □□□□□□ URL
```

□□□□ 94 (□□: `10-ultimate-guide.md`)

```
Notion □□□□□□□ → n8n Webhook □□
  → □□ Coze Agent API → □□□□
  → □□ MP4
  → □□□□□
    ├── TikTok API
    ├── YouTube Shorts API
    └── Instagram Reels API
```

□□□□ 95 (□□: `10-ultimate-guide.md`)

```
git clone https://github.com/AIDC-AI/Pixelle-Video
# □□ LLM □□□GPT/DeepSeek/Ollama/□□□□□
# □□□□□□□□□
# □□□□□□□□□□□□□□□□□□□□□□□
```

□□□□ 96 (□□: `10-ultimate-guide.md`)

```
// Claude Desktop MCP □□□□□Kling AI□
{
  "mcpServers": {
    "kling-ai": {
      "command": "node",
      "args": ["path/to/kling-mcp/index.js"],
      "env": { "KLING_ACCESS_KEY": "...", "KLING_SECRET_KEY": "..." }
    }
  }
}
```

□□□□ 97 (□□: `10-ultimate-guide.md`)

```
models/diffusion_models/
  └── wan2.2_i2v_high_noise_14B_fp8_scaled.safetensors  (Image-to-Video 14B)
models/clip/ □ models/text_encoders/
  └── umt5_xxl_fp8_e4m3fn_scaled.safetensors
models/vae/
```

```
  └── wan2.2_vae.safetensors
models/loras/
  └── wan2.2_t2v_lightx2v_4steps_lora_v1.1_high_noise.safetensors  (□□ LoRA)
```

□□□□ 98 (□□: `10-ultimate-guide.md`)

```
□□□□□ 2 □□□□□□□□□ $5-20
□□□□60 □
□□□□□□□~$100□Runway Std + ElevenLabs + CapCut□
□□□□$300-1200
ROI□200-1100%
```

□□□□ 99 (□□: `11-one-click-best-practices.md`)

```
MidJourney □□□ → Runway Gen-4.5 □□□□□□□□□□
                   □ Seedance 2.0 □□□□□□□□□□5-8 □/□□
                   □ Veo 3.1 □□□□□□□□□/□□□□□□□
→ ElevenLabs □□ → CapCut Pro □□+□□ → Artlist □□□□
```

□□□□ 100 (□□: `11-one-click-best-practices.md`)

```
□□□LLM□→ □□□AI □□ or □□□□→ □□□TTS□→ □□□ASR□→ □□ → □□□FFmpeg□
```

□□□□ 101 (□□: `11-one-click-best-practices.md`)

```
□□□□□□□□□□□□□□□□  □□□□ — Pixelle-Video □ n8n □□
□□□□/□□□□□□        □□□□ — □□□□□ + □□□□ + □□□□
□□□□/□□□□          □□ — Runway □□ + Veo □□ + Premiere □□
```

□□□□ 102 (□□: `11-one-click-best-practices.md`)

```
① Claude (claude.ai) / GPT-4o (chat.openai.com) → □□□□□□ JSON □□□
② Flux 2 (via fal.ai) / MidJourney (midjourney.com) → □□□□□□□□□□□□□□
□ □□□□□□□□□□□□
   ├── □□/□□/□□ → Runway Gen-4.5 (runwayml.com) — Motion Sketch □□□□
   ├── □□□□/□□ → Seedance 2.0 (dreamina.capcut.com) — @ □□□□
   ├── □□/□□/□□□□ → Veo 3.1 (aistudio.google.com) — □□□□
   └── □□/□□/□□ → Kling 3.0 (klingai.com) — Motion Control
④ ElevenLabs (elevenlabs.io) → □□□□□ Veo □□□□□
⑤ CapCut (capcut.com) / Premiere → □□+□□+□□
⑥ Suno (suno.com) / Artlist (artlist.io) → □□
```

□□□□ 103 (□□: `11-one-click-best-practices.md`)

```
Google Sheet □□□□
  → n8n Schedule Trigger
```

```
  → ChatGPT/Claude → □□ JSON
  → PiAPI(Flux) → □□□
  → PiAPI(Kling) → Image-to-Video
  → ElevenLabs → □□
  → Creatomate → □□□□□□
  → OpenAI → □□□□
  → Blotato/upload-post → □□□□□ 5 □□□
  → Discord Webhook → □□
```

□□□□ 104 (□□: `11-one-click-best-practices.md`)

```
ComfyUI □□□□
  ├── FLUX 2 → □□□□□
  ├── Wan 2.1 14B (via WanVideoWrapper) → Image-to-Video□□□□□
  │   □ LTX-2 (via ComfyUI-LTXVideo) → Image-to-Video□□□□□□
  ├── Frame Interpolation → □□□ 60fps
  ├── NVFP4 □□ → 60% □□□□
  ├── RTX Video Super Resolution → □□ 4K □□
  └── Export → MP4/WebM/ProRes
```

□□□□ 105 (□□: `11-one-click-best-practices.md`)

```
git clone https://github.com/AIDC-AI/Pixelle-Video
# □□ LLM□GPT/DeepSeek/Ollama/□□□□□
# □□□□□□□FLUX/Qwen□+ □□□□□Wan 2.1□
# □□□□ → □□□□□→□□→□□→□□→BGM→□□
# Windows □□□□□□□
```

□□□□ 106 (□□: `11-one-click-best-practices.md`)

```
□□ 5s □□ → □□□□□□ → Topaz □□ (topazlabs.com) → □□□□□ I2V □□ → □□□ 5s → □□
```

□□□□ 107 (□□: `11-one-click-best-practices.md`)

```
Prompt: "...No background music. Only natural ambient sounds and dialogue."
```

□□□□ 108 (□□: `11-one-click-best-practices.md`)

```
@Image1□□□□□□+ @Video1□□□□□□□□+ @Audio1□□□□□□+ □□□□
```

□□□□ 109 (□□: `11-one-click-best-practices.md`)

```
□□ 1 → □□□□ 1 □ → □□□□ 2 □□□ → □□□□ 2 → □□□□ 1 □ → ...
```

□□□□ 11O (□□: `11-one-click-best-practices.md`)

```
                    ┌──────────────────────────────┐
                    │     □□□□□□□□□□          │
                    │  n8n (n8n.io)                │
                    │  Make (make.com)             │
                    │  Coze (coze.cn)              │
                    └──────────────────────────────┘
                                │
                                │
            ┌───────────────────┼────────────────────┐
            │                   │                    │
    ┌───────▼──────────┐   ┌──▼──────────┐   ┌──▼──────────────┐
    │ □□□□□ComfyUI□│  │ API □□□   │  │ Agent □□       │
    │ + WanVideoWrapper │  │ Kie.ai      │  │ Luma Agents (□!)│
    │ + LTX-2.3        │  │ fal.ai      │  │ Higgsfield      │
    │ + Kie-API □□    │  │ Abacus.AI   │  │ Coze Agent      │
    │ + PiAPI          │  │ PiAPI       │  │                 │
    └──────────────────┘   └─────────────┘   └─────────────────┘
            │                   │                    │
            └───────────────────┼────────────────────┘
                                │
            ┌───────────────────▼────────────────────┐
            │         □□□□□□□              │
            │  □□□Kling 3.0 (klingai.com)  │
            │        Veo 3.1 (aistudio.google)│
            │        Runway 4.5 (runwayml.com)│
            │        Seedance 2.0 (dreamina)  │
            │        Grok Imagine (x.ai) (□!) │
            │  □□□Wan 2.1 (GitHub)         │
            │        LTX-2 (GitHub)           │
            │        SkyReels-V2 (GitHub)     │
            └──────────────────────────────────┘
```